TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (VPA DD)

PUBLICATION DATE **29.06.2023**

VERSION **v.2.3**

RELATED SUPPORT

- Programme of Activity requirements
- TEMPLATE GUIDE VPA Design Document

This document contains the following sections

KEY PROJECT INFORMATION

Type of VPA	☐ Real case VPA ☒ Regular VPA
Scale of VPA Note that a VPA can be of one scale. Please select applicable scale accordingly.	☒Microscale ☐Small scale ☐Large scale
Title of corresponding real case VPA (if applicable)	GS11993 VPA1: African Clean Energy for Cooking in Uganda
GS ID of real case VPA (if applicable)	GS11994
GS ID of VPA	GS12817
Title of VPA	GS11993 GS11994 VPA3: African Clean Energy for Cooking in Uganda
Time of First Submission Date	30/04/2024
Date of Design Certification	
Version number of the VPA-DD	Version 4
Completion date of version	03/11/2024
Coordinating/managing entity	African Clean Energy BV
VPA Implementer (s)	ACE Energy Solutions Uganda Limited
Project Participants and any communities involved	ACE Energy Solutions Uganda Limited
Host Country (ies)	Uganda
GS ID and Title of applicable Design Certified VPA	GS11993 VPA1: African Clean Energy for Cooking in Uganda
GS ID and Title of applicable Performance Certified VPA	
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities

	☐ N/A
Other Requirements applied	N/A
Methodology (ies) applied and version number	Gold Standard "Methodology for metered & measured energy cooking devices"[1], version 1.2
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
VPA Cycle:	☒ Regular ☐ Retroactive

Land-use & Forest and Agriculture - Key Project Information[2]

Table 1 – Estimated Sustainable Development Contributions

SUSTAINABLE DEVELOPMENT GOALS TARGETED	SDG IMPACT (DEFINED IN B.6.)	ESTIMATED ANNUAL AVERAGE	UNITS OR PRODUCTS
13 Climate Action (mandatory)	Reduced GHG emissions	7,153	t CO_2
3 Good health and wellbeing	• Reduced (or avoided) kitchen smoke exposure • Reduced reported health impacts of kitchen smoke	• 2,081 • 1,755	person-years person-years
5 Gender Equality	Time saved by women faster cooking	377,849	hours saved per year in total
7 Affordable and clean energy	Access to improved cooking technology	9,401	person-years
8 Decent work and economic growth	Employment generation	40	full-time jobs

[1] https://globalgoals.goldstandard.org/430_ee_ics_methodology-for-metered-measured-energy-cooking-devices/

[2] Please refer to 0 for detailed information on LUF projects

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

Under this VPA, ACE distributes micro gasifier cookstoves in Uganda. ACE primarily utilizes a B2C model, ensuring we bring our customers a holistic experience directly to their homes. Not only can our customers use agricultural waste or small sticks around their home, they can also choose to have our sustainable briquettes delivered directly to their door. This helps with one of our primary goals, helping transition our customer base off charcoal use. However, charcoal can also still be used in the ACE One as a transitionary fuel.

We also provide fast and friendly after-sales service to all our customers, even after the two-year warranty period finishes. Our service includes at least one of our team members going to the customer, ensuring they can continue cooking as soon as possible.

This project will start utilizing the latest version of the ACE One energy system, which will now be produced in Uganda.

The project boundary includes physical, geographical sites of stove usage and baseline and project fuel production.

Tentative VPA implementation plan

The project implementation plan can be summarized as follows:

1. Q2 2024 – Q2 2025: Approx. 2,500 units produced in Uganda distributed and sold across Uganda
2. Q3 2025 – First Monitoring Campaign of this VPA

A.1.1. Eligibility of the VPA under approved PoA

Table 2 Eligibility for VPA inclusion as per PoA requirements

NO.	ELIGIBILITY CRITERION	DESCRIPTION/ REQUIRED CONDITION	DESCRIPTION OF THE VPA IN RELATION TO THE CRITERIA, MEANS OF VERIFICATION AND SUPPORTING EVIDENCE FOR INCLUSION
1	The Geographical boundaries of VPAs are consistent with the geographical boundary of the PoA	The geographical boundary of the VPA is within the geographical boundary of the PoA	The VPA is within Uganda, as shown in section A.2.
2	Conditions to avoid double counting of Impacts, such as unique identifications of product and end user locations	A unique ID will be assigned to each customer in each VPA, and each device will receive a unique serial number; moreover, user locations will be registered by coordinates.	Unique identification systems described under the VPA section B.7.3.
3	Conditions to confirm that VPAs are neither registered as project activities with other offset Schemes, included in other registered PoAs, nor the project activities that have been deregistered;	The VPA, nor any of its devices, is not yet registered and not being registered as a standalone project under other carbon standards by ensuring that the VPAs has the full title over the emission reductions generated by the users listed in the VPA.	There is a mechanism to ensure the transfer of legal ownership to the emission reductions to the CME, as described in section A.1.2. A declaration is provided excluding double-counting.
4	Specification of the technology/measure such as the level and type of service, as well as performance specification based on, inter alia, testing/certification;	VPAs under this PoA will consist in the distribution of improved cook stoves to users cooking primarily with non-renewable biomass on traditional biomass stoves (mainly charcoal stoves) in the baseline scenario. The improved cook stove technology will have a thermal efficiency of at least 40%.	As described in section A.3, the stove model used (ACE One) is described in the PoA-DD in detail.

5	Conditions to check the start dates of V/CPAs through documentary evidence;	A start date will be specified for each VPA. All VPAs will have the start date after the start date of the PoA.	As shown in C.1.1, the start date of the VPA is 01/05/2023, after the start day of the PoA.
6	Conditions to ensure compliance with the applicability of the applied methodologies, the applied standardized baselines and the other applied methodological regulatory documents	Each VPA will meet all the applicability criteria of the Gold Standard "Methodology for metered & measured energy cooking devices", version 1.2, as explained in par. 2.2 of the methodology.	Section B.2 shows the that the inclusion criteria for methodology application are met.
7	Conditions to ensure that V/CPAs meet the requirements for demonstration of additionality	Each VPA will demonstrate automatic additionality according to the GS community services activity requirements section 4.1.9. (c) – Microscale projects. As the PoA and VPAs are fall under microscale projects, hence deemed additional.	Section B.5 shows automatic additionality.
8	Conditions to ensure no diversion of official development assistance;	There will be no diversion of ODA for any of the proposed VPAs	A declaration of non-use of ODA is provided, and a corresponding statement is made in section A.5.
9	Target group (e.g. domestic/commercial/industrial, rural/urban, gridconnected/offgrid), and where applicable, distribution mechanisms (e.g. direct installation)	The PoA will target charcoal- and/or wood dependent households based in urban, semi-urban and rural, areas within the host countries. Also fuelwood/charcoal-dependent institutions may be targeted. Project devices will be installed directly in the participating households.	Section A.3 shows that the criterion is met.
10	Conditions related to sampling requirements for the PoA	Conditions are in line with the GS "Methodology for metered & measured energy cooking devices", version 1.2 (Sampling may not be required since 100% fuel consumption is measured for SDG 13).	B.7.2 explains the sampling approach (non-systematic sampling, only required for a qualitative SDG 3 parameter).

11	Conditions to ensure that VPAs that will be included meet the small-scale or microscale thresholds and remain within those thresholds throughout the crediting period (N/A if all units qualify as "microscale CDM units")	N/A	N/A
12	Conditions to confirm that technologies in V/CPAs are eligible	Covered by inclusion criterion 4	Covered by inclusion criterion 4.
13	Conditions to be met by each VPA regarding SDG outcomes assessment	Positive outcomes expected for at least 3 SDGs.	This is the case, as shown in section B.6.4
14	(if applicable) Conditions to be met by each VPA regarding safeguarding principles	The safeguarding principles assessment is conducted at VPA level.	As shown in Annex 1, no specific conditions are derived from safeguarding principles assessment.
15	Eligibility as per Community Services Activity Requirements	As described in the assessment at PoA level (see PoA-DD A.3), specific criteria to be met at VPA-level are covered by eligibility criterion No. 4 (technology includes end-use energy efficiency) and legal ownership (clear description of ownership, proofs that end users are aware and willing to give up rights on products) and discussion of the transfer of ownership during LSCs.	Eligibility of technology covered by inclusion criterion 4. Conditions on legal ownership: - Clear description in A.1.2 LSC report showing discussion on transfer of ownership.

16	Eligibility as per GHG Emissions Reduction & Sequestration criteria	The relevant criteria include: - Only microscale VPAs are allowed under a microscale PoA (criterion 2.1.3) - There is no mandatory operational scheme to reduce GHG emissions (3.1.1) - The VPA belongs to an eligible project type: End-use energy efficiency improvement (5.1.1.b)	- A.4 shows that this VPA is a microscale VPA - No mandatory scheme exists in Uganda[3] - A.3 shows that technology consists in efficient cookstoves, so the VPA belongs to the type of End-use energy efficiency improvement
17	Conditions to be met for retroactive VPAs	The start date of retroactive VPAs must be within one year of the first submission of the PoA.	N/A
18	Conditions to be met for a VPAs in another country than Uganda (condition for multi-country PoA)	The VPA must apply a similar technology; it must consist in the distribution of advanced cooking technologies eligible under the "Methodology for metered & measured energy cooking devices", and it must meet the eligibility criteria of this methodology. Moreover, a physical stakeholder consultation must be organized in cases where required according to PoA requirements (see section E of PoA-DD).	N/A

[3] The most recent NDC does not emission trading schemes or market-based approaches for GHG reduction. On page 12 it says: "Uganda will use voluntary cooperation provided for in Article 6 in accordance with the National Climate Change Act 2021 to demonstrate her mitigation ambition and mobilise support to promote sustainable development and poverty eradication. **Non-market approaches** such as Adaptation Benefit Mechanism will be explored among others. This shows that currently, there are no plans to establish such a scheme in the near future. https://unfccc.int/sites/default/files/NDC/2022-06/Uganda%20interim%20NDC%20submission_.pdf

A.1.2. Legal ownership of products generated by the VPA and legal rights to alter use of resources required to service the project

The legal ownership of the emission reductions is with ACE. The same was discussed during the LSC meeting. ACE signs agreements with all the beneficiaries confirming the transfer of all legal rights of the emission reductions.

Producers of briquettes or other project fuels such as pellets will provide statements that legal ownership of carbon credits generated with ACE One stoves is with ACE.

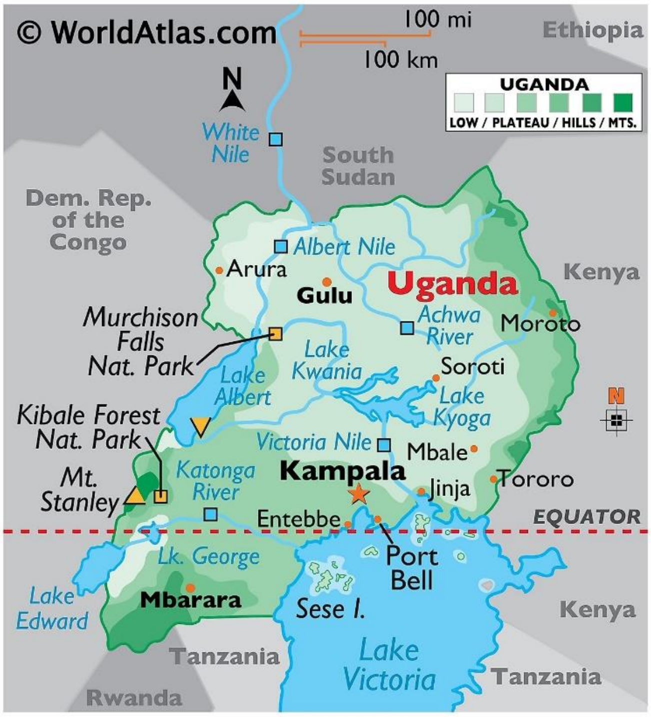
A.2. Location of VPA

Country: Uganda

The geographical coordinates of the VPA are (Kampala):

Latitude: 0°18.9768′ N

Longitude: 32°34.9314′ E



A.3. Technologies and/or measures

The ACE One is a solar/biomass hybrid energy system that is significantly more efficient than traditional cooking methods, while also providing the ability to light someone's home, to charge a phone, and to reduce health concerns. The latter is made possible through a smokeless cooking experience thanks to gasification technology. ACE One works with a huge variety of fuels. Users can use agricultural waste or small sticks around their home, they can also choose to have our sustainable briquettes delivered directly to their door. This helps with one of our primary goals, helping transition our customer base off charcoal use. However, charcoal can also still be used in the ACE One as a transitional fuel.

The target group includes households and small institutions such as restaurants that are primarily fuelwood/charcoal-dependent.

The stove's useful cooking power is up to 1.5kW (see CREEC ISO efficiency test protocol).

The estimated life time of the ACE One is 5 years[4], but this is just a minimum estimate. ACE is in regular contact with all its customers, including a fast and friendly after-sales service, even after the two-year warranty period finishes. Our service includes at least one of our team members going to the customer. Maintenance and replacement of spare parts assure a long lifetime, and if a stove cannot be repaired, it is replaced. In cases where a stove is not replaced or stops being used for another reason, it is not accounted for emission reductions anymore, due to the continuous 100% usage monitoring.
Stove efficiency is 50.9%. No decrease of efficiency is expected because in case of failures, pyrolysis for gasification and thereby stove operation will completely stop.

[4] A statement of the stove producer is provided, explaining about the minimum lifetime of 5 years and the regular maintenance and exchange of spare parts that are conducted under this PoA and that allow for even longer lifetime.

Moreover, regular maintenance will assure that the stoves are kept in a good state. Maintenance evaluation shows that there is no reduction ins air flow over time[5].

Positive contributions to SDGs are as follows:
SDG 13, Climate Action: Reduction of GHG emissions by replacing non-renewable cooking fuels (mainly charcoal) with highly efficient biomass briquettes and by increasing combustion efficiency where charcoal is still used.
SDG 3, Good health and wellbeing: Avoidance of negative health impacts due to smoke from conventional charcoal stoves – the project stoves burn with a clean flame comparable to LPG.
SDG 5, Gender Equality: The project stoves are much faster to light than conventional charcoal stoves, saving time particularly to women who mostly do the cooking.
SDG 7: Affordable and clean energy: Clean gasifier stoves are made available to customers; due to the high efficiency, customers can reduce cooking fuel expenses.
SDG 8, Decent work and economic growth: The project generates employment in stove distribution and in the briquette supply chain, among others.

A.4. Scale of the VPA

The project is microscale as the stoves that will be used in this project are expected to represent an annual CO2 reduction of up to 10,000 tonnes of CO2e.

A.5. Funding sources of VPA

The project is financed by ACE. There is no ODA funding involved in the implementation of this VPA that comes under the condition of delivering carbon credits to a donor.

[5] Analysis provided to SustainCert

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

The project applies the Gold Standard Methodology: Gold Standard "Methodology for metered & measured energy cooking devices"[6], version 1.2. The methodology will be referred to as "MECD" in this document.

B.2. Applicability of methodology (ies)

Applicability Criterion as in 2.2 of MECD	Justification of applicability
a) Project shall choose a technology design that has predictable performance in that it is proven to be efficient and durable under field conditions; for fuel-based cookstoves, the rated thermal efficiency shall be higher than the baseline technology and at least 40%.	The certified thermal efficiency is 50.9% for the ACE One stove (certificates provided). ACE One stove has proven to be efficient and durable over many years. The main baseline technology are wood and charcoal stoves with efficiencies of 10% and 20%.
b) The technology shall have continuous useful energy output of less than 150kW per unit, where "continuous useful energy output" is defined (MECD 1.1.1 a.) as the energy transferred to the contents of a cooking vessel, including the sensible heat that raises the temperature of the contents of the cooking vessel and the latent heat of evaporation of	The maximum useful energy output of the ACE One stove is in the range of 1.5kW, as shown in the different ISO test provide for the determination of thermal efficiency. The highest value

6 https://globalgoals.goldstandard.org/430_ee_ics_methodology-for-metered-measured-energy-cooking-devices/

water from the cooking vessel, divided by the time of the operation of the cooking task.	found is 1,475kW (phase 2 of charcoal ISO test).
c) The project activity is implemented by a project developer and can include additional project participants listed in Appendix 2 of the PDD template. The individual households and institutions may be represented collectively by community organisations, etc., but do not individually act as project participants.	It is confirmed that individual households and other stove users will not act as project participants.
d) The project developer must design incentive mechanism(s), which should be effective as fast as possible, for the elimination of any inefficient baseline stoves that are replaced entirely by the project cooking device(s) and describe the incentive mechanism(s) in the PDD/VPA-DD at the time of validation.	The most effective incentive is the cost advantage of cooking with briquettes on the ACE One stove in comparison to baseline stoves. For charcoal usage, the cost advantage is evident by the increased efficiency of the ACE One in comparison to baseline stoves (45.9% vs. 20%) which directly translates into a cost advantage due to lower charcoal consumption. For briquettes, eh advantage is still higher, due to an even higher stove efficiency of 50.9%. There is also an advantage of briquettes over charcoal to be used in the ACE One. The cost per cooking energy is lower[7],

[7] Briquettes are sold at 40,000 UGX per 40kg bag, and a 15% discount for regular data synchronizing. With a stove efficiency of 50.9% and an NCV of 16.2MJ/kg, this results in a cost of 26-31UGX per MJ useful cooking energy. Charcoal is sold at 80,000-160,000 UGX per 40kg bag (*Charcoal price hike leaves many in pain - Uganda (africa-press.net)*). With a stove efficiency of 45.9% and an NCV of 29.5MJ/kg, the resulting cost per MJ useful cooking energy is in average 46UGX.

	and delivery is to the users' home in clean bags with accurate packaging sizes.
e) To avoid double counting or double claiming, the project developer must: i. clearly communicate its ownership rights and intention of claiming the emission reductions resulting from the project activity to the following parties by contract or clear written assertions in the transaction paperwork: all other project participants; project technology manufacturers; and retailers of the project technology or the renewable fuel in use; and ii. inform and notify the end users that they cannot claim emission reductions from the project or use of devices distributed as part of the project activity, and iii. exclude from the project activity, project devices included in any other voluntary market or CDM project activity/PoA, and strive not to displace the cooking technologies of another CDM or voluntary project/PoA. See Section, 3.11, avoidance of double counting or double claiming with other mitigation actions, for details on this demonstration.	i. The transfer of ownership on carbon credits has been discussed during the LSC and it is confirmed by each individual user in the user agreement. Fuel producers provide specific statements, the technology provider is identical with the CME. ii. The same is also confirmed in the user agreement iii. This is ensured by a unique identification number of each household receiving a stove under this project (See B.7.3). Moreover, a declaration on double counting is provided.
f) Under this methodology, emission reductions cannot be claimed for fuel-switch only, so proposed project activities also need to introduce alternative technologies, i.e. technology switch is also involved.	The project introduces micro gasifier stoves, a new technology.
g) For project cooking devices that use fossil fuel, only emission reductions from efficiency improvement are eligible.	Project devices use biomass.
h) For project cooking device that use grid electricity, emission reductions from fuel switch and efficiency improvement are eligible.	No grid electricity is used for cooking. Stoves have a fan for which electricity is used from a

	solar panel attached to the stove. Surplus electricity can be used for lighting or phone charging, thereby even replacing grid electricity.
i) The measured fuel or energy is used to calculate both baseline and project emissions. The project developer must have monitoring systems in place to monitor the fuel or energy consumption by all the project devices under the project to be recorded in a database, which is maintained by the project developer.	Fuel consumption of all project devices will be monitored continuously and thoroughly (see B.7.1). At the same time, fan running time is registered and will be used to calculate emission reduction for those times where the stove is running on other fuels than the briquettes delivered.

Additional note on the eligibility according to the scope of the of the technology as "metered energy cooking device"

The definition for metered cooking devices in 1.1.1c requires devices to "record fuel or energy use directly, or <u>through a supplementary meter with the ability to record amount of energy or fuel used for cooking over a period of time.</u>
The ACE One meets this definition in two ways:

1. The amount of sustainable fuel (briquettes) delivered to stove users is recorded. This fuel switch approach is similar to the case of ethanol cookers mentioned under 1.1.1c, including *"… ethanol cookers <u>when metered and sold for use in a dedicated device</u>*", which apparently refers to metered ethanol fuel, in analogy to briquettes. Briquettes usage without the ACE One stove is not plausible, as explained in B.7.2.
2. Fan running time is recorded automatically by each stove; it is equivalent to stove usage time and thus to fuel usage. It is read out regularly. Where fan running times are registered that are not (fully) explained by briquette consumption, additional ER are generated and claimed through usage of the ACE One with the primary baseline fuel, leading to a significant fuel savings by efficiency gain.

B.3. VPA boundary

The VPA boundary is defined in accordance to 3.1 of the RECH methodology v.4.0

a. The project boundary is the physical, geographical sites of the project technologies/practices including the fuel collection and production area.

   i. Where the baseline fuel is woody biomass (including charcoal), the project boundary also includes the area within which this woody biomass is grown and collected.

The boundary includes all of Uganda, including the market for biomass fuels.

   ii. For projects using processed fuels, this boundary also includes the baseline and project fuel production (e.g. charcoal, plant oil) and solid waste and effluents disposal or treatment facilities associated with fuel processing.

The project uses sustainable biomass briquettes produced in Uganda, within the VPA boundary. Emissions from briquette production are included under project emissions. Also charcoal production plant are within the VPA boundary.

   iii. In cases where the project activity introduces the use of a new biomass feedstock into the project situation, the fuel production and collection area is the area within which this new biomass is produced, collected and supplied.

The project uses sustainable biomass briquettes produced in Uganda, within the VPA boundary. Emissions from briquette production are included under project emissions.

	Source	GHGs	Included?	Justification/ Explanation
Baseline scenario	Delivery of thermal energy from charcoal, wood and some LPG for cooking, and also very marginal amounts of other fuels such as electricity.	CO_2	Yes	Important source of emissions
		CH_4	Yes	Important source of emissions
		N_2O	Yes	Minor source of emissions
	Production of fuel, transport of fuel	CO_2	Yes	Important source of emissions
		CH_4	Yes	Important source of emissions
		N_2O	Yes	Minor source of emissions
Project scenario	Delivery of thermal energy from briquettes and charcoal	CO_2	Yes	Minor source of emissions
		CH_4	Yes	Minor source of emissions
		N_2O	Yes	Minor source of emissions
	Production of briquettes from residual and possibly also non-renewable biomass and transport (electricity and fuel consumption)	CO_2	Yes	Minor source of emissions
		CH_4	Yes	Minor source of emissions
		N_2O	Yes	Minor source of emissions

B.4. Establishment and description of baseline scenario

In absence of the project, households and institutions cook on traditional stoves with conventional fuels, mainly charcoal (the project targets users with charcoal as primary fuel). A systematic third-party baseline survey was conducted by CIRCODU, commissioned by Clan Cooking Alliance[8]. Based on this study, typical fuel mixes are determined for two user groups, primary charcoal users and primary firewood users. LPG is hardly used; averages are considered for all users as a conservative assumption. Users who primarily use LPG are not considered for emission reductions and other impact calculation.

As required by MECD 3.4.1, a baseline emission factor per useful energy unit is derived from the mix of baseline fuels for each of the user groups which is then applied to the useful cooking energy delivered by the project.

B.5. Demonstration of additionality

Note on additionality:

The 'clean cooking' business has historically shown itself to be difficult to make profitable, given the impact considerations, and the substantial cost of last-mile distribution. Most companies in this business would no longer exist had it not been for various kinds of grant funding, and this funding isn't sizeable or predictable enough to bring enterprises to scale. Carbon finance serves to enable ACE to sustainably work with low gross profit margins on hardware that keep the customer and context central. Furthermore, ACE has since 2015 offered its customers interest-free credit terms to pay off its systems over a period of up to 12 months, which is costly.

On the fuel side, ACE injects (within certain parameters, to avoid perverse incentives) a significant portion of (future) carbon revenue into the price of sustainable biomass pellets and briquettes to ensure not just affordability, but affordability to the point where customers are able to procure in bulk. This is absolutely necessary if we are to disincentivise the usage of unsustainable charcoal and firewood.

The ACE One differs from the vast majority of cheap rocket stove products in this sector, in terms of quality and cost, and ACE is proud to manufacture in the markets it operates, including Uganda, but also Lesotho and Cambodia.

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version	According to the GS community services activity requirements section 4.1.9. (c) – community services microscale projects are automatically additional.

[8] A final draft is available at time of VPA submission, shared with SustainCert

number and the specific paragraph, if applicable).	
Describe how the proposed VPA meets the criteria for deemed additionality.	The VPAs falls under microscale projects and GS community services activities (see eligibility criterion 15), hence it is deemed additional.

B.5.1. Prior Consideration

Milestones showing prior consideration:

- May 20 2022: Physical meeting of the Local Stakeholder Consultation held, for the real case VPA-DD GS11994, including a discussion on the PoA GS11993 with the clear intention of including several VPAs.
- June 29 2022: Start of distribution of ACE One cookstoves under GS11994
- May 19st 2024: Start of distribution of ACE One cookstoves under this VPA, under similar conditions.

B.5.2. Ongoing Financial Need

N/A

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

SUSTAINABLE DEVELOPMENT GOALS TARGETED	MOST RELEVANT SDG TARGET	SDG IMPACT **INDICATOR (PROPOSED OR SDG INDICATOR)**
13 Climate Action	13.2 Integrate climate change measures into national policies, strategies and planning	Reduced GHG emissions

3 Good health and wellbeing	3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	• Avoided kitchen smoke exposure Reduced reported health impacts of kitchen smoke
5 Gender equality	5.4 Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	Time saved by women from faster cooking
7 Affordable and clean energy	7.1 By 2030, ensure universal access to affordable, reliable and modern energy services	Access to improved cooking technology
8 Decent work and economic growth	8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Employment generation

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

Baseline Emissions

According to the methodology MECD, a baseline emission factor per unit of useful cooking energy will be established, based on the fuel mix in the pre-project case and baseline stove efficiencies. Baseline emissions will then be calculated by multiplying the useful cooking energy provided by project devices (based on monitoring of briquette consumption) with the baseline emission factor.

<u>Baseline Emission Factor:</u>

$$EF_{b,useful} = \sum_k \left(\sum_{i,j} P_{b,i,j} \times Percentage\ of\ fuel_i \times (EF_{b,i,CO2} \times (fNRB_{i,y} + EF_{b,i,non-CO2})_k \times NCV_{b,i,j}\right) \div \sum_k \left(\sum_{i,j} P_{b,i,j} \times\times Percentage\ of\ fuel_i \times NCV_{b,i} \times \eta_{b,i,j}\right)_k$$

Eq 1 of MECD

Where:

$EF_{b,useful}$	=	Baseline emissions factor (tCO_2e per TJ of useful energy), for entire mix of baseline cooking energy
$P_{b,i,j}$	=	Amount of baseline fuel *i* used in device *j* in year *y* (tonnes)
$EF_{b,i,CO2}$	=	Emission factor of the baseline fuel *i* (tCO_2e/tonne)
$EF_{b,i,non-CO2}$	=	Emission factor of the baseline fuel *i* (tCO_2e/tonne)
$fNRB_{i,y}$	=	Non-renewability status of woody biomass fuel i during year y
$NCV_{b,i}$	=	The net calorific value of the baseline fuel type *i* (TJ/tonne)
$\eta_{b,i,j}$	=	Efficiency of baseline device *j* with fuel i (fraction)
$Perentage\ of\ fuel\_i_i$	=	Percentage of fuel type I in baseline situation (%)
k	=	Household *k* from the target population
j	=	Baseline devices *j*
i	=	Baseline fuel *i*

Parameter values for Eq. 1 are obtained as described in the parameter tables (B.6.2). On $\boldsymbol{P_{b,i,j}}$ explanations are also given here below. On ***fNRB***, explanation have been moved to the bottom of this section B.6.1., for better readability.

$\boldsymbol{P_{b,i,j}}$

Baseline fuels considered include charcoal, firewood and minor amounts of LPG[9]. When signing the contract, users indicate which fuel is used primarily, and the corresponding baseline profile is applied. Users who primarily use LPG will not be considered under this VPA. *fNRB* is applied for charcoal, but obviously not for LPG.

Overall baseline emissions are calculated as per Eq. 2:

$$BE_y = EG_{p,useful,y} \times EF_{b,useful}$$ *Eq. 3 of MECD*

Where:

BE_y	=	Baseline emissions (tCO_2e) in the year y
$EG_{p,useful,y}$	=	The amount of useful energy applied in the project in year y (TJ)
EF_b	=	Baseline emissions factor (tCO_2e per TJ of useful energy)

The amount of useful energy applied in the project is derived by Eq. 4. (Eq. 3 of MECD is not applicable since no electric stoves are deployed):

$$EG_{p,useful,y} = \sum_{d,i} P_{p,i,d,y} \times NCV_{p,i} \times \eta_{p,d,y}$$ *Eq. 7 of MECD*

Where:

$P_{p,i,d,y}$	=	The amount of fuel i used in the project in by device *d* in year *y*, considering cap (mass or volume unit)
$NCV_{p,i}$	=	The net calorific value of the fuel *i* used in the project scenario in year *y*
$\eta_{p,d,y}$	=	Energy efficiency of the project device, *d* in year *y* (fraction)
d	=	Project device *d*

[9] Other fuels such as electricity may be used very marginally during the rainy season, but are neglected for the baseline calculation. As shown in the ER calculation sheet under "baseline emission factor", there is a clear increase of charcoal usage in the wet season which is not considered in the baseline calculation, for conservativeness. This conservative approach outweighs by far the neglection of marginal usage of other fuels in the rainy season.

All parameters used to calculate Eq. 4 are monitored, as explained in B.7.1 and B.7.3.

$P_{p,i,d,y}$ is measured consecutively in two steps:

a) Records of briquette sales to ACE One users show the quantity of briquettes consumed ($P_{p,briquettes,y}$)

b) Fan running time is recorded. In cases where no briquettes have been consumed, or where some briquettes have been consumed, but fan running time shows additional usage of the stove, the quantity of the baseline fuel consumed is calculated for the additional ("surplus") fan running time. As a conservative assumption, the baseline fuel is always assumed to be charcoal – the "dirtiest" baseline fuel. Useful cooking energy is thus calculated from the surplus fan running time with assumed charcoal consumption[10] ($P_{p,baselinefuel,y}$).

The corresponding formula is:

$$P_{p,baselinefuel,d,y} = (Fantime_{d,y} - P_{p,briquettes,d,y} * rate_{cons,briquettes}) * rate_{cons,baselinefuel}$$

Where:

$Fantime_{d,y}$ = Time of operation of stove d (hours)

$P_{p,briquettes,d,y}$ = Briquette consumption of stove d (kg)

$rate_{cons,briquettes}$ = Consumption rate of briquettes in the ACE One stove (kg/h)

$rate_{cons,baselinefuel}$ = Consumption rate of the primary baseline fuel (always charcoal assumed) in the ACE One stove (kg/h)

$EG_{p,useful,y}$ is thus derived by summarizing useful cooking energy from briquettes and baseline fuels in the ACE One for different fuels, and the corresponding $NCV_{p,i}$ is applied to each of them.

[10] Fan running time can only be read out with a connection to a smartphone. For stove with no or incomplete data on fan running time, only briquette consumption is used to calculate ER.

Project Emissions

A first source of project emissions is the combustion of briquettes or the baseline fuel in the project stove.

Briquettes are mostly made from sawdust which is waste biomass, equivalent to renewable biomass. Where raw biomass other than sawdust is used that cannot be deemed waste biomass, project emissions are calculated by applying the same factors that apply for wood consumed in the baseline.

$$PE_{combustion,\ briquettes,y} = P_{p,briquettes,total,y} * conversion_{wood\_briquettes} * share_{non\text{-}waste} * (EF_{b,wood,CO2} * f_{NRB} + EF_{b,wood,non\text{-}CO2})$$

Where:

$PE_{combustion,\ briquettes,y}$	= Project emissions caused by burning briquettes in the project stove
$conversion_{wood\_briquettes}$	= Wood-to-briquette conversion factor (1.2, meaning that 1.2kg dry woody biomass are required to produce 1t of dry briquettes)
$share_{non\text{-}waste}$	= Percentage of the raw material not deemed waste biomass (equivalent to non-renewable), monitored.
$EF_{b,wood}$	= Emission factors for wood (see B.6.2)

Where the baseline fuel is consumed in the ACE One (charcoal always assumed conservatively), PE are also derived by applying the emissions factors used in the baseline.

$$PE_{combustion,\ charcoal} = P_{p,charcoal,total,y} * (EF_{b,charcoal,CO2} * f_{NRB} + EF_{b,charcoal,non\text{-}CO2})$$

Since the project uses processed biomass-based fuel (briquettes), project emissions associated with production and transport of fuel must be evaluated (3.6.2 of MECD).

Briquette production uses electricity. A conservative value of 100kWh/ton is used, based on data from the briquette factory[11]. The grid emission factor is set conservatively as 0.6tCO2/MWh, according to the CDM standardized baseline[12].

[11] Provided to SustainCert

[12] https://cdm.unfccc.int/methodologies/standard_base/2015/sb45.html It shows combined margins between 0.48 and 0.58, and a build margin of 0.454, showing a decreasing tendency. A more recent standardized baseline under https://cdm.unfccc.int/methodologies/standard_base/2015/sb176.html shows a much lower grid factor, but the factor from VPA1 is kept for reason of conservativeness and uniformity between VPAs.

Project emissions from electricity consumption are thus calculated as (according to equation 1 of CDM tool 5 version 3):

$$PE_{electricity,y} = 0.1\ MWh * P_{briquettes,y} * 0.6tCO_2/MWh\ (1+20\%) = 0.072tCO2 * P_{total,y}$$

Where:

$PE_{electricity,y}$ = Project emissions caused by electricity consumption for briquette production

$P_{total,y}$ = Total amount of briquettes consumed in project devices in a monitoring period

Transport emission have to be considered according to 3.6.3 C of MECD, if distances are over 200km. Most users will be within a distance below that threshold. At verification, it is aimed to show that the average distance is <200km. If this cannot be shown, for all users in distance >200km, transport emissions will be are considered by applying an emission factor given in the CDM tool "Project and leakage emissions from transportation of freight, version 1.1"[13]. In section 5.3, a default value of 129 gCO2/tkm is given for heavy vehicles.

$$PE_{transport,l,y} = 0.129kg/CO2 * distance_l * P_{distance,l,y}$$

Where:

$PE_{transport,l,y}$ = Project emissions caused by transport of briquettes to location l

$Distance_l$ =distance between briquette factory and location l

$P_{distance,l,y}$ = Total amount of briquettes consumed in location l

$$PE_y = PE_{combustion,briquettes,y} + PE_{combustion,charcoal,y} + PE_{electricity,y} + PE_{transport,y}$$

[13] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-12-v1.1.0.pdf

Leakage Emissions

MECD makes reference to 3.11 of RECH. Here an approach is described for a systematic assessment of leakage.

a. The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or with a higher intensity than would have occurred in the absence of the project.

Baseline stoves will often remain in use, while continued usage is automatically considered when monitoring fuel consumption in the project stove. If baseline stoves should be displaced, it is not plausible that other households would use these devices and in consequence increase biomass consumption. Since biomass is the cheaper option, non-availability of conventional biomass stoves is not a limiting factor.

b. Members of the population who do not participate in the project, and previously used lower emitting energy sources, instead use the non-renewable biomass or fossil fuels saved under the project activity.

The VPA is too small to have a significant effect on the charcoal market, given that there are some 10 million households in Uganda.

c. The project significantly reduces the NRB fraction within an area where other GHG mitigation project activities account for NRB fraction in their baseline scenario.

The influence of the VPA on fNRB is insignificant, given that there are some 10 million households in Uganda.

d. The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of space heating or by retaining some use of inefficient technology.

The climate in Uganda is very stable, average temperatures are >20° in nearly all the country. Moreover, baseline stoves will often remain in use, while continued usage is automatically monitored by metering briquette consumption.

e. By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution with this technology by households who commonly used a technology with relatively lower emissions.

All users are known to the project, both stoves and fuel are sold centrally. Moreover, the baseline survey considers average usage of different technologies (of which the project technology is still the most efficient).

In conclusion, no leakage emissions have to be considered.

Calculation of Emission Reductions

$$ER_y = BE_y - PE_y - LE_y$$ *Eq. 10 of MECD*

Where:

ER_y	=	Emission reductions in year y (t CO2e/yr)
BE_y	=	Baseline emissions in year y (t CO2e/yr)
PE_y	=	Project emissions in year y (t CO2/yr)
LE_y	=	Leakage emissions in year y (t CO2/yr)

Determination of the fraction of non-renewable biomass

As per the CDM tool "Calculation of the fraction of non-renewable biomass" (V.4.0)[14]. The overall equation to be applied is:

(equation 1 of tool)

$$fNRB = \frac{NRB}{NRB + RB}$$

where:

fNRB: *Fraction of non-renewable biomass (fraction or %)*

NRB: *Quantity of non-renewable biomass (t/yr)*

RB: *Quantity of renewable biomass (t/yr)*

The quantity of non-renewable biomass is derived by the equation given by:

$$NRB = H - RB$$ (***equation 2 of tool)***

Where:

H: Total annual consumption of wood in the absence of the project activity in tons/yr

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[14] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf

<u>Determination of the total wood consumption *H*</u>

$$H = HW * N + CE + NE - RB \qquad \textbf{\textit{(equation 3 of tool)}}$$

Where:

*HW * N:*	*Total household's fuelwood consumption in tons. The tool par. 15 allows to use an aggregated value for consumption per household (HW) times the number of households (N).*
CE:	*Commercial woody biomass consumption for energy applications (e.g., commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)*
NE:	*Commercial woody biomass consumption for non-energy applications (e.g. construction, furniture) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)*

The detailed fNRB calculation is shown in the corresponding Excel sheet.

The result:

$$fNRB = 81\%$$

This lower than in other cookstove projects in Uganda[15].

According to Bailis et al. (2015)[16], Uganda is part of a hot spot in Eastern Africa (see also figure 4 of the article). The map presented by Bailis et al. (2015) shows values up to 80% for Uganda. The value of 81% seems plausible in the light of high importance of charcoal among the VPA's target group. In difference to firewood, charcoal is mostly produced from standing trees; since it is easier to transport, it is often brought from relatively dry areas where hard wood is found. It has thus a higher impact on forests than firewood which is often collected from dead biomass.

Methodological choices for non-carbon SDGs:

These are mentioned under B.6.3, for the sake of simplicity and uniformity with the VPA-DD of the real case VPA.

15 89% approved in the recent GS10734 GS1247 VPA 255 Kaliro Safe Wate ttps://platform.sustain-cert.com/public-project/2116

16 https://www.nature.com/articles/nclimate2491/

B.6.2. Data and parameters fixed ex ante

SDG13

Data/parameter ID	MECD 1
Data/parameter	$P_{b,i,j}$
Data unit	Tonne/ year-device
Description	Amount of baseline fuel i used in device j (tonnes) in the baseline.
Source of data	A very recent study on baseline fuel usage for cooking in Uganda is used, prepared by CIRCODU and commissioned by Clean Cooking Alliance. A final draft is provided to SustainCert.
Value(s) applied	Primary charcoal users: $P_{b,firewood}$ 0.81 t/a $P_{b,charcoal}$ 0.99 t/a $P_{b,LPG}$ 0.0011 t/a Primary firewood users $P_{b,firewood}$ 4.23 t/a $P_{b,charcoal}$ 0.19 t/a $P_{b,LPG}$ 0.0011 t/a
Choice of data or Measurement methods and procedures	Third party study. For details see ER calculation sheet.
Purpose of data	Emission reduction calculation
Additional comment	

Data/parameter ID	MECD 2
Data/parameter	$NCV_{b,i}$
Data unit	Terrajoules (TJ)/tonne of fuel (charcoal and LPG)
Description	The net calorific value of the baseline fuel type i
Source of data	IPCC default data: 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol 2 Chapter 1, Table 1.2
Value(s) applied	Wood: 0.0156 Charcoal: 0.0295 LPG: 0.0473
Choice of data or Measurement methods and procedures	IPCC default values
Purpose of data	Emission reduction calculation
Additional comment	n/a

Data/parameter ID	MECD 3
Data/parameter	$EF_{b,i,CO2}$
Data unit	tCO_2e/TJ
Description	CO_2 emission factor arising from use of fuels in baseline scenario
Source of data	Wood: Methodology default Charcoal: Methodology cap LPG: IPDD default (Table 2.5 of Chapter 2 of Vol. 2, 2006 IPCC Guidelines)
Value(s) applied	Wood: 112 Charcoal: 197.15

	LPG: 63.1
Choice of data or Measurement methods and procedures	**Charcoal:** The cap given in MECD is applied because it still represents a conservative estimate. As shown below, realistic emission factors for Uganda would still be much higher than the cap, due to low conversion efficiency in charcoal plants. This is shown by an analysis of official UNdata data, comparing hypothetical wood-to-charcoal conservation factors that are not applied under this VPA: The amount of fuelwood transformed in charcoal plants is indicated as 27,933 *1000m3[17]. If applying the FAO default density factor of 0.725t/m3[18], this corresponds to 16,525 tons. Charcoal production is indicated as 2,849 * 1000t[19]. This results in a wood-to-charcoal conversion factor of 5.8 (16,525/2,849). However, the density value applied is by far underestimated because for charcoal production, heavy woods are preferred. **Theoretically, a wood-to-charcoal conversion clearly above 6 would thus be well justified.** The corresponding theoretical wood-to-charcoal factor from **MECD cap values is only 4.5**, calculated from: [EFcharcoal,CO2 (197.15tCO2/TJ) + EFcharcoal,non-CO2 (92.29tCO2/TJ)]* NCVcharcoal (0.0295TJ/t) / [EFwood,CO2 (112tCO2/TJ) + EFwood,non-CO2 (9.46tCO2/TJ)]* NCVwood (0.0156 TJ/t) = 4.5 This comparison of hypothetical wood-to-charcoal conversion factors shows that applying the cap of the charcoal EF is still a conservative choice.
Purpose of data	Emission reduction calculation
Additional comment	The parameter is used to calculate baseline emissions

[17] data.un.org/Data.aspx?q=fuelwood&d=EDATA&f=cmID%3aFW
[18] http://www.fao.org/docrep/010/a1106e/a1106e00.htm
[19] data.un.org/Data.aspx?q=charcoal&d=EDATA&f=cmID%3aCH

Data/parameter ID	MECD 4
Data/parameter	$EF_{b,i,nonCO2}$
Data unit	tCO_2e/TJ
Description	Non-CO_2 emission factor arising from use of fuels in baseline scenario
Source of data	Wood: Methodology default, AR5 GWP Charcoal: Methodology cap, AR5 GWP LPG: n.a.
Value(s) applied	Wood: 9.46 Charcoal: 92.29
Choice of data or Measurement methods and procedures	**Charcoal:** See explanation under parameter MECD 3 above.
Purpose of data	Emission reduction calculation
Additional comment	The parameter is used to calculate baseline emissions

Data/parameter ID	MECD 5
Data/parameter	$\eta_{b,i,j}$
Unit	Fraction
Description	Energy efficiency of baseline device *j* with fuel *i*
Source of data	Charcoal: MECD default value for other conventional systems using woody biomass LPG: Scientific paper [20]
Value(s) applied	Charcoal: 20% Wood: 10%

[20] cited on http://aprovecho.org/the-big-picture/lpg-cooking-and-health

	LPG: 51%
Choice of data or Measurement methods and procedures	- According to the methodology for metered & measured cooking devices: Other conventional systems using woody biomass: default efficiency 20%
Purpose of data	Emission reduction calculation
Additional comment	The parameter is used to determine the baseline emission factor.

Data/parameter ID	MECD 6
Data/parameter	*Percentage of fuel_i*
Unit	%
Description	Percentage of fuel type I in baseline situation.
Source of data	From the baseline study, two baseline user profiles are identified, consuming different shares of multiple fuels (see parameter table for $P_{b,i,j}$). Therefore, it would be redundant to apply a percentage again to these shares. The parameter is thus set to 100%.
Value(s) applied	100%
Choice of data or Measurement methods and procedures	See explanation under source of data.
Purpose of data	Emission reduction calculation
Additional comment	n/a

Data/parameter ID	No ID assigned, not described in MECD
Data/parameter	$rate_{cons,i}$
Unit	Kg/h
Description	Consumption rate of fuel i in the ACE One stove
Source of data	See separate calculation sheet.
Value(s) applied	$\text{rate}_{\text{cons,briquettes}}$ = 0.64 kg/h $\text{rate}_{\text{cons,charcoal}}$ = 0.39 kg/h
Choice of data or Measurement methods and procedures	Detailed explanation in separate excel sheet. Based on an evaluation of fuel consumption rates during ISO test phases, compared with data on power levels used under field conditions.
Purpose of data	Emission reduction calculation
Additional comment	n/a

Data/parameter ID	No ID assigned, not described in MECD
Data/parameter	$Conversion_{wood\text{-}briquettes}$
Unit	t/t
Description	Conversion factor wood-briquette. This parameter is only needed if non-renewable biomass is used to produce briquettes, i.e. if $Share_{non\text{-}waste}$ >0 (see B.7.1). It indicates how much woody biomass is used as input to generate a ton of briquettes.
Source of data	Derived from factory data.
Value(s) applied	1.2

Choice of data or Measurement methods and procedures	When woody biomass is transformed into briquettes, a mass reduction occurs, basically due to moisture reduction. Incoming biomass typically has a moisture content between 20% and 45%, since it is stored in huge piles (for example, sawdust at sawmills). Briquettes have an average moisture content around 5%. During moisture reduction, the carbon content of biomass does not change. For the calculation of project emission due to combustion of briquettes made from non-renewable biomass ($PE_{combustion,briquettes,y}$), default values for NCV_{wood} and EF_{wood} for air dried wood with up to 20% moisture are used[21]. If solely considering differences in moisture, 1kg of briquettes corresponds to 1.14 kg wood (120%/105%). Normally, all biomass is transformed into briquettes. However, there may be some losses of biomass in the process, due to dust and unusable bark (which is waste in anyway). We conservatively assume a loss of 6% biomass. The resulting wood-to-briquette factor is thus set as 1.2
Purpose of data	Project Emission reduction calculation
Additional comment	Fixed for the project for the crediting period. It will be reassessed at each crediting period renewal.

[21] AMS-II.G parameter table "NCVwood"; 20% fuelwood moisture is a conservative estimate for air dried wood: https://uk-air.defra.gov.uk/assets/documents/reports/cat09/1903131256_Seasoning_Wood_Web_Feb_2019_V5.pdf ; IPCC Guidelines for GHG inventories Vol. 2 Ch.2 table 2.9: *"Values were originally based on gross calorific value; they were converted to net calorific value by assuming that net calorific value for dry wood was 20 per cent lower than the gross calorific value."*

Data/parameter ID	No ID assigned, entirely based on other Parameters
Data/parameter	$EF_{b,useful}$
Unit	tCO_2e per GJ of useful energy
Description	Baseline emissions factor (for entire baseline cooking energy mix)
Source of data	Calculated according to equation 1 of MECD (see B.6.3 and ER calculation sheet for the detailed calculation).
Value(s) applied	Emission factor applicable to primary charcoal users: $EF_{b,useful,charbase}$ = 1.21 Emission factor applicable to primary wood users: $EF_b, woodbase$ = 1.04
Choice of data or Measurement methods and procedures	See other parameter tables in this section and calculation in B.6.3, as well as ER calculation sheet
Purpose of data	Emission reduction calculation
Additional comment	Fixed for the project for the crediting period. It shall be reassessed at each crediting period renewal. This parameter table was included due to the central position of $EF_{b,useful}$

B.6.3. Ex ante estimation of SDG Impact

SDG 13 Reduced GHG emissions

A step-by-step calculation according to formulas in the MECD methodology is shown below. This calculation is applied in the ER calculation sheet.

Step 1: Baseline emission factor

$$EF_{b,useful} = \sum_k \left(\sum_{i,j} P_{b,i,j} \times Percentage\ of\ fuel_i \times (EF_{b,i,CO2} \times (fNRB_{i,y} + EF_{b,i,non-CO2}\right)_k \times NCV_{b,i,j}) \div \sum_k \left(\sum_{i,j} P_{b,i,j} \times Percentage\ of\ fuel_i \times NCV_{b,i} \times \eta_{b,i,j}\right)_k$$

See detailed calculation in ER calculator in Excel

Step 2: Useful energy delivered by the project

$$EG_{p,useful,y} = \sum_d P_{p,d,y} \times NCV_{p,i} \times \eta_{p,d,y}$$

See detailed calculation in ER calculator in Excel

Step 3: Baseline emissions derived from useful project energy

$$BE_y = EG_{p,useful,y} \times EF_{b,useful}$$

See detailed calculation in ER calculator in Excel

Step 4: Emission reductions

$$ER_y = BE_y - PE_y - LE_y$$

See detailed calculation in ER calculator in Excel

SDG 3: Avoided or reduced smoke exposure

$$SmokeAvoided_y = N_{improvedstove,y} * Share\_woodcharcoal\_baseline$$

Where:

$SmokeAvoided_y$	Avoided kitchen smoke exposure in person-years
$N_{improvedstove,y}$	Number of smoke-reduced cookstoves (such as portable rocket stoves)
$Share\_woodcharcoal\_baseline$	Share of charcoal and wood energy usage in the baseline

Preliminary annual result per stove: $SmokeAvoided_y$ = 1 stove * 99.6% of charcoal and wood usage in the baseline (from MP1 of GS11994) = 1 person-year

SDG 3: Reduced health impacts

$$Healthimprovement_y = N_{P,y} * Share_{healthimproved,y}$$

Where:

$Healthimprovement_y$	Reported health improvements such a cough reduction in person-years
$Share_{healthimproved,y}$	Share of positive respondents (ex-ante estimate). Will be monitored.

Preliminary annual result per stove: $Healthimprovement_y$ = 1 stove * 84% of positive respondents (from MP1 of GS11994) = 0.84 person-years

SDG 5: Time savings

For ex-ante calculations:

$$Timesavedtotal_y = N_{p,y} * Timesavedstove * Share_{female\_cooks,y}$$

Where:

$Timesavedtotal_y$	Accumulated time saved from faster cooking
$Timesavedstove$	Average time saved per stove due to faster cooking (mainly stove lighting)
$Share_{female\_cooks,y}$	Conservative estimate for the share of women cooking

Preliminary annual result per stove: $Timesavedtotal_y$ = 1 stove * 1.09h * 365days * 45% = 179 hours

Note: The share of female cooks was derived from the share of cookstove purchase contract partners in MP1 of GS11994 which underestimates the share of female cooks.

SDG 7: Improved access to cleaner cooking energy

$$Accesscleanercooking_y = N_{P,y} * Householdsize$$

Where:

$Accesscleanercooking_y$	Access to clean cooking in person-years
$Householdsize$	Average number of persons per household

Preliminary annual result per stove: $Acccesscleanercooking_y$ = 1 stove * 4.5 = 4.5 person-years

SDG 8: Job openings)

Calculated based on payrolls. Standardized to person-years of fulltime employment (for example, two persons with half time job are equivalent to one person with a full-time job.

Preliminary annual average: 40 person years (40 full-time employee equivalents)

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 13: Reduced GHG emissions

Results are based on the distribution of 2,500 stoves with an average ER for 3.54tCO2/a (preliminary assumption). ER measured in tCO2e

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2024 (from 19.05. on)	4,790	1,213	3,576
2025	9,580	2,427	7,153
2026	9,580	2,427	7,153
2027	9,580	2,427	7,153
2028	9,580	2,427	7,153
2029 (till 18.05.)	4,790	1,213	3,576
Total	47,898	12,135	35,763
Total number of crediting years: 5*			
Annual average over the crediting period	9,580	2,427	7,153

*half of 2024 and half of 2029 considered

SDG 3: Reduced kitchen smoke exposure

Results are based on 2,500 households targeted. Calculated in person-years.

Assumption: A household switches from 80% charcoal cooking on a traditional charcoal stove to cooking on a project stove. This household contributes 0.8 person-years of reduced smoke exposure per year.

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2024 (from 19.05. on)	0	1,040	1,040
2025	0	2,081	2,081
2026	0	2,081	2,081
2027	0	2,081	2,081

2028	0	2,081	2,081
2029 (till 18.05.)	0	1,040	1,040
Total		10,404	10,404
Total number of crediting years: 5*			
Annual average over the crediting period		2,081	2,081

*half of 2024 and half of 2029 considered

SDG 3: Reduced reported health impacts of kitchen smoke

Results are based on 2,500 households targeted.

Calculated in person-years. Assumption: 80% of households report perceived health improvements. An average household therefore contributes 0.8 person-years of reduced smoke exposure per year.

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2024 (from 19.05. on)	0	877	877
2025	0	1,755	1,755
2026	0	1,755	1,755
2027	0	1,755	1,755
2028	0	1,755	1,755
2029 (till 18.05.)	0	877	877
Total		**8,744**	**8,744**
Total number of crediting years: 5*			
Annual average over the crediting period		1,755	1,755

*half of 2024 and half of 2029 considered

SDG 5: Time saved from faster stove lighting procedure

Results are based on 2,500 households targeted. Calculated in total hours saved, assuming 10min saved per day.

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2024 (from 19.05. on)	0	188,924	188,924
2025	0	377,849	377,849
2026	0	377,849	377,849
2027	0	377,849	377,849
2028	0	377,849	377,849
2029 (till 18.05.)	0	188,924	188,924
Total		**1,889,243**	**1,889,243**
Total number of crediting years: 5*			
Annual average over the crediting period		377,849	377,849

*half of 2024 and half of 2029 considered

SDG 7: Access to clean cooking

Results are based on 2,500 households targeted, with average household size of 4.5[22]. (Access to clean cooking is calculated per person). Calculated in person-years, accounting for all household members (4.5 assumed).

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2024 (from 19.05. on)	0	4,700	4,700

[22] www.ubos.org/wp-content/uploads/publications/09_2021Uganda-National-Survey-Report-2019-2020.pdf
This source indicates a value of 4.6. However, 4.5 is applied for conservativeness.

2025	0	9,401	9,401
2026	0	9,401	9,401
2027	0	9,401	9,401
2028	0	9,401	9,401
2029 (till 18.05.)	0	4,700	4,700
Total		**47,003**	**47,003**
Total number of crediting years: 5*			
Annual average over the crediting period		9,401	9,401

*half of 2024 and half of 2029 considered

SDG 8: Job Openings

Calculated in person-years. Preliminarily, 40 full-time employees assumed annually.

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2024 (from 19.05. on)	0	20	20
2025	0	40	40
2026	0	40	40
2027	0	40	40
2028	0	40	40
2029 (till 18.05.)	0	20	20
Total		200	200
Total number of crediting years: 5*			
Annual average over the crediting period		40	40

*half of 2024 and half of 2029 considered

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Data/parameter ID	MECD 9
Data / Parameter	$\eta_{p,d,y}$
Unit	Fraction
Description	Thermal efficiency of the project device with briquettes and charcoal.
Source of data	Certified ISO tests from CREEC laboratory (certificates provided)
Value(s) applied	Efficiency with briquettes ($\eta_{p,briquettes,y}$): 50.9% Efficiency with charcoal ($\eta_{p,charcoal,y}$): 45.9% No decrease of these value is considered (see below).
Measurement methods and procedures	Certified ISO tests (certificates provided)
Monitoring frequency	Annual
QA/QC procedures	Regular maintenance of stoves as described to ensure that ensure full functioning of the critical factors decisive for efficiency, see A.3.
Purpose of data	This parameter is used in the determination of useful energy
Additional comment	Non-decrease of efficiency is plausible since in case of failures, pyrolysis for gasification and thereby stove operation will stop, see A.3.

Data/parameter ID	MECD 13
Data/parameter	$f_{NRB,i,y}$
Unit	Fraction non-renewability
Description	Non-renewability status of woody biomass fuel in scenario i during year y
Source of data	Calculated according to CDM tool 30 version 4
Value(s) applied	81%
Measurement methods and procedures	See details in B.6.1 and fNRB sheet in Excel.
Monitoring frequency	Ex-ante for the first crediting period
QA/QC procedures	Latest version of the CDM tool 30 used (version 4)
Additional comment	Will be reassessed at renewal of crediting period.

Data/parameter ID	MECD 14a
Data / Parameter	$P_{briquettes,d,y}$
Unit	tonnes
Description	The amount of wood briquettes used in the project in device d in year y, considering cap
Source of data	Direct measurement by briquettes sales.
Value(s) applied	Preliminary estimate: 180 kg/a per device (15kg per month)
Measurement methods and procedures	Briquette sales data to customers, mostly on an individual base, partly on a clustered basis.
Monitoring frequency	Continuously, aggregated monthly
QA/QC procedures	Measurement occurs by evaluating sales receipts for packaging units with standardized content. Users pay for briquettes, therefore exhaustive usage is not plausible. Briquettes are exclusively distributed by the project and

	there are no sensible alternative uses for briquettes; they do not burn well in other cookstove stoves and would be far too expensive for re-selling to biomass boilers etc. A corresponding statement by an independent research institution is provided. Packaging is done with calibrated equipment according to industry standards and relevant national requirements. For calculating ER, a cap will be applied to individual energy consumption corresponding to a per-capita energy consumption of 0.0045 GJ per day. In order to avoid crediting of unused briquette at the end of a monitoring period, only transactions taking place after the start of the monitoring period will be considered, and in addition, no ER will be claimed for the first 7kg provided to each new user after. This has been added under B.7.3.
Purpose of data	Emission reduction calculation
Additional comment	

Data/parameter ID	MECD 14b
Data / Parameter	$P_{baselinefuel,d,y}$
Unit	tonnes
Description	The amount of baseline fuel used in the project in device d in year y, considering cap Baseline fuel is always assumed to be charcoal, for conservativeness.
Source of data	Measurement through monitoring of surplus fan running time not covered by briquette consumption.
Value(s) applied	Preliminary estimate: Charcoal: 124 kg/a per device, based on 319 hours surplus fan running time at a consumption rate of 0.39kg/h (see ER calculation sheet for details)
Measurement methods and procedures	The ACE One stove automatically recorded fan running time. It has to be read out with a smartphone.

	The formula applied is: $P_{p,baselinefuel,d,y} = (Fantime_{d,y} - P_{p,briquettes,d,y} * rate_{cons,briquettes}) * rate_{cons,baselinefuel}$ Where: $Fantime_{d,y}$ = Time of operation of stove d (h) $P_{p,briquettes,d,y}$ = Briquette consumption of stove d (kg) $rate_{cons,briquettes}$ = Consumption rate of briquettes in the ACE One stove (kg/h) $rate_{cons,baselinefuel}$ = Consumption rate of the baseline fuel (charcoal assumed) in the ACE One stove (kg/h)
Monitoring frequency	Continuously, aggregated monthly
QA/QC procedures	For calculating ER, a cap will be applied to individual energy consumption corresponding to a per-capita energy consumption of 0.0045 GJ per day.
Purpose of data	Emission reduction calculation
Additional comment	The baseline fuel is always assumed to be charcoal, even for primary baseline wood users, for conservativeness.

Data/parameter ID	No ID assigned, not described in MECD
Data / Parameter	$Fantime_{d,y}$
Unit	Hours (but derived in seconds from the stove)
Description	Time of operation of stove d
Source of data	Automatically recorded by the ACE One stove. It has to be read out with a smartphone and is then submitted to the central database.
Value(s) applied	Preliminary estimate: 50hours/month and stove in average
Measurement methods and procedures	The ACE One stove automatically records fan running time. It has to be read out with a smartphone. This is either done by stove users or by ACE Uganda.

Monitoring frequency	Recorded continuously in real time. Read out and submitted to the database occasionally, generally every few weeks.
QA/QC procedures	Where fan times appear very low or high, ACE Uganda will try to get explanations from users.
Purpose of data	Emission reduction calculation

Data/parameter ID	MECD 15
Data / Parameter	LE_y
Unit	tCO_2e per year
Description	Leakage in project scenario in year y
Source of data	Option 2 of MECD is followed: Evaluate leakage following the procedure described there.
Value(s) applied	0 (no discount)
Measurement methods and procedures	No significant potential sources of leakage have been identified (see B.6.2). The assessment will be repeated at least annually.
Monitoring frequency	Annually
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Emission reduction calculation
Additional comment	Emissions from briquette production and transport are considered as Project Emissions.

Data/parameter ID	No ID assigned, not described in MECD
Data / Parameter	$NCV_{p,i}$
Unit	Terrajoules (TJ)/tonne of fuel
Description	The net calorific value of briquettes type i

Source of data	Analyses in independent laboratories
Value(s) applied	0.0162
Measurement methods and procedures	ISO test certificate CREEC (thermal efficiency testing ACE One with briquettes)
Monitoring frequency	Annually
QA/QC procedures	Analyses in independent laboratories, cross-check with comparable measurements of other briquettes.
Purpose of data	Emission reduction calculation
Additional comment	

Data/parameter ID	No ID assigned, not described in MECD
Data/parameter	$Share_{non\text{-}waste}$
Unit	%
Description	Indicates the percentage of biomass used for briquette production that cannot be deemed waste biomass, i.e., that is regarded non-renewable biomass. For this share, project emissions are calculated as for wood in the baseline.
Source of data	Monitored
Value(s) applied	10%, preliminarily. (Normally, all briquettes are produced from sawdust that is waste biomass.)
Measurement methods and procedures	Statement of briquette producer
Monitoring frequency	Annually
QA/QC procedures	
Purpose of data	Emission reduction calculation

Non-carbon SDGs

Note: Indicators with standardized units such as "person-years" are used. These units have the advantage of representing absolute (not relative) values, therefore being summable and making projects comparable between each other. (For details, see https://www.fairclimatefund.nl/content/1-onze-aanpak/5-onze-impact/guidance-project-level-sdg-indicators-27mar2021.pdf).

Data / Parameter	SDG 3: $SmokeAvoided_y$:
Unit	Person-years of cooking with avoided smoke exposure
Description	Accumulated life time of ICS users during which they are able to cook with avoided smoke exposure from wood and charcoal stoves.
Source of data	Monitoring campaigns
Value(s) applied	Preliminary value: 2,081 person-years of cooking with reduced smoke exposure, per year (based on values from MP1 of GS11994).
Measurement methods and procedures	Calculated from the share of total cooking done with the ACE One stove in replacement of wood and charcoal stoves. Only one person assumed to cook per household. Example: A household switches from 80% charcoal cooking on a traditional charcoal stove to cooking on the Ace One stove. This household contributes 0.8 person-years of reduced smoke exposure per year.
Monitoring frequency	Annually
QA/QC procedures	Certificate on emissions from ACE One stove.
Purpose of data	Calculation of project scenario
Additional comment	The baseline smoke exposure is not measured physically, instead an average smoke exposure from traditional cookstoves is assumed. If possible, accompanying studies directly measuring smoke concentration (PM, CO, …) might be used to substantiate the findings.

Data / Parameter	SDG 3: $Healthimproved_y$: Reduced reported health impacts of kitchen smoke
Unit	Person-years with reported health improvement
Description	Accumulated life time of ICS users during which they are report improved health due to avoided exposure to smoke from charcoal stoves.
Source of data	Monitoring campaigns
Value(s) applied	Preliminary value: 1,755 person-years of cooking with reduced smoke exposure, per year, assuming 84% of positive respondents (value from MP1 of GS11994).
Measurement methods and procedures	Calculated from the total number of stove users time the share of respondents indicating health improvements ($Share_{healthimproved,y}$). Only one person assumed to cook per household.
Monitoring frequency	Annually
QA/QC procedures	Certificate on emissions from ACE One stove.
Purpose of data	Calculation of project scenario
Additional comment	The parameter is based on subjective user perceptions. If possible, accompanying studies directly measuring smoke concentration (PM, CO, …) might be used to substantiate the findings.

Data / Parameter	SDG 5: $Timesavedtotal_y$: Total time saved from faster cooking
Unit	Total accumulated hours saved by woman using ACE One.
Description	Time savings achieved by women who can reduce cooking times due to faster lighting of the ACE One.
Source of data	Stove database, fuel sales showing stove usage.
Value(s) applied	1.09 hours per day (value from MP1 of GS11994).
Measurement methods and procedures	Calculated based on ACE One stove usage and survey data on time consumption on baseline stoves.
Monitoring frequency	Annually
QA/QC procedures	
Purpose of data	Calculation of project scenario
Additional comment	

Data / Parameter	SDG 7: $Accesscleanercooking_y$:
Unit	Person-years of using clean cookstoves
Description	Accumulated life time of persons in households during which the clean ACE One stove is used instead of a charcoal stove.
Source of data	Monitoring
Value(s) applied	Preliminary value: 9,401 (based on values from MP1 of GS11994).
Measurement methods and procedures	Calculated from the share of total cooking done with the ACE One stove in replacement of charcoal stoves. Example: A household of 5 switches to cooking on the ACE One stove. This household contributes 5 person-years of clean cooking per year.
Monitoring frequency	Annually
QA/QC procedures	
Purpose of data	Calculation of project scenario
Additional comment	Calculated per household member, since SDG focuses on clean energy access, not smoke reduction.

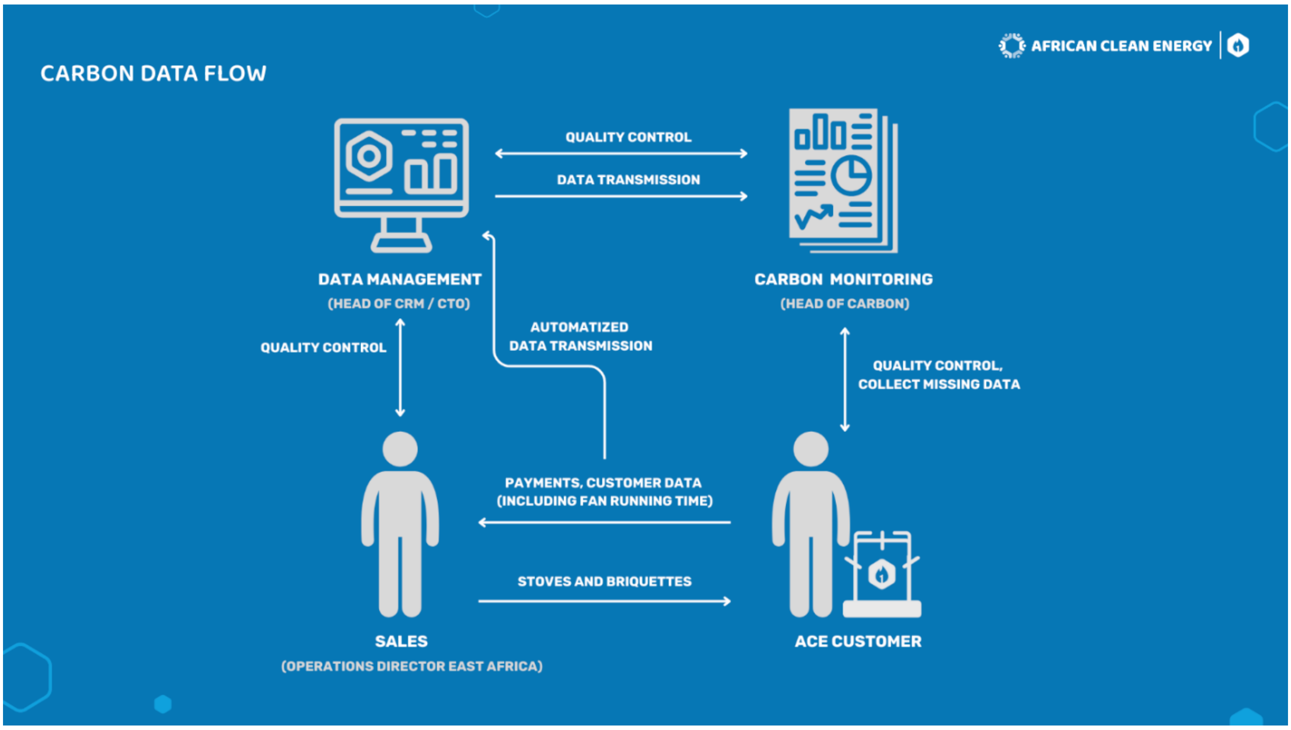
Data / Parameter	SDG 8: $Jobopenings_y$ Employment generation by the project
Unit	Person-years of employment due to the project
Description	Employment generation standardized to person-years of fulltime employment.
Source of data	Monitoring
Value(s) applied	Preliminary value: 40 (third part of the value from MP1 of GS11994 – impact split over three VPAs).
Measurement methods and procedures	Calculated based on payrolls. Standardized to person-years of fulltime employment (for example, two persons with half time job are equivalent to one person with a full-time job.
Monitoring frequency	Annually
QA/QC procedures	
Purpose of data	Calculation of project scenario
Additional comment	

B.7.2. Sampling plan

The only variables that are not monitored continuously through stove distribution and briquette sales are $Healthimproved_y$, the perceived effects of smoke reduction, and time saved from faster cooking. Since both are rather qualitative variable, non-systematic sampling will be applied. Perceptions of at least 30 users will be gathered, for example, during a defined period before verification, users that get in touch with project staff for fuel purchases or stove maintenance will be asked about perceived health impacts and time savings.
Alternatively, phone interviews will be conducted with at least 30 randomly selected users. This number is deemed sufficient because both are qualitative parameters that do not require high statistical significance.

B.7.3. Other elements of monitoring plan

Plan for monitoring data management



In addition, a detailed presentation on monitoring data management has been provided to the VVB.

Regular maintenance monitoring

For each stove, the system records maintenance events which are ordered by customers through the user APP. Maintenance is therefore fully monitored.

A. Unique numbering system

All customer data are managed in an electronic database (details given below under B). Two central unique identifiers are used (the current practice below might be adjusted in the future, but always maintaining uniqueness):

A unique **Stove serial number** is generated by the system (Salesforce) and engraved to each stove. It is then included both in the user contract and in the database, serving as number of the "stovebook" where all relevant information is assigned to the stove and the customer using it. It is thereby also used to link customer data and briquette purchases, among others.

-

B. Stove Deployment Record:

The VPA implementer will maintain and continuously update an electronic database with all stove records. The data management system currently used is Salesforce. The main purpose of the database is to track customer payments for stoves and briquettes, but at the same time, it is suitable to monitor data relevant for carbon calculation.

The following data are recorded (adjustments possible):

i. Customer ID
ii. Stove serial number
iii. Date of stove installation
iv. User name
v. Location of household
vi. Telephone number (if available)
vii. GPS coordinates of the households (if available)
viii. Primary baseline fuel used
ix. Briquette purchases (regularly updated)
x. Fan hours recorded (regularly updated)
xi. Maintenance records

Data entries are created when inserting user data in a smartphone app. Each user also signs a paper contract when obtaining the ACE One, containing the data saved in the database, except contract reference and GPS data which are automatically assigned by the system when inserting user data. Paper contracts are stored and serve as a backup. Data on the primary baseline fuel are obtained from a separate survey conducted when a user joins. Paper contracts may be replaced by electronic contracts in the future.

C. Briquette sales record

Briquette sales to customers are recorded continuously by the data management system. Briquette sales come with personalized purchase records since every payment is tied to the unique stove serial number (and thus customer and stove).

A detailed presentation on monitoring data management has been provided to the VVB.

There may also be cases where briquettes are purchased cluster-wise. For example, corporate customers may buy briquettes bulk-wise for their employees who are stove owners with personally registered stoves. It will always be made clear how end-users belong to a cluster.

Briquettes are packed into standardized packages, generally of 7kg (provided to new users) and 40kg (packaging sizes might be changed in the future). Packaging occurs with calibrated equipment.

Measures to guarantee that briquettes are exclusively used in project stoves

At validation stage, it is confirmed that there are no sensible alternative usages for wood briquettes by ACE One users in Uganda than usage in the ACE One stove, because:

- Wood briquettes are not suitable as a fuel in other common cookstoves
- Reselling briquettes for other potential usages, such as in biomass boilers, would make no economic sense for ACE One users, because they would receive much lower prices than they paid.
- Briquette sales to ACE One users are tracked in a personalized manner. Misuse would be easily detected.

Together with each monitoring event, these points will be evaluated again.

A statement on exclusive usage is provided to the VVB signed by an independent expert entity (CREEC laboratory of Makerere University in Kampala).

Moreover, it will be shown in each monitoring event that $P_{p,briquettes,y}$ exclusively refers to briquettes that are used in stoves included under this VPA. This will be done by:

- Considering only briquette purchases taking place after the start of the monitoring period, and not claiming ER for the first 7kg used by a new user.
- Investigating any individual user's briquette consumption that surpasses the MECD cap of 0.0045GJ per capita and day, to either find a transparent and plausible explanation (large household, he may order briquettes for his neighbor also), or to consider only the amount below the cap for ER calculation.

D. Evaluation of fan running time

The ACE One stove automatically records fan running time. It has to be read out with a smartphone. This is either done by stove users or by ACE Uganda. When read out, data are automatically stored in the database. It is read out occasionally, generally every few weeks. However, there are cases where fan running time is not read out, if, for example, the user does not possess a smartphone and is not visited by ACE staff. In such cases, no ER will be claimed for surplus cooking time.

If briquette consumption is less than required to explain fan running time, charcoal is assumed to be used during the surplus cooking time (see B.6.1 and parameter tables in B.7.1).

However, there are cases where fan running time is not read out, if, for example, the user does not possess a smartphone and is not visited by ACE staff. In such cases, no ER will be claimed for surplus cooking time.

A detailed presentation on monitoring data management has been provided to the VVB.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of VPA

19/05/2024

C.1.2. Expected operational lifetime of VPA

20 Years

C.2. Crediting period of project

C.2.1. Start date of crediting period

19/05/2024

C.2.2. Total length of crediting period

5 Years

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in Appendix 1, ongoing monitoring is summarised below.

Safeguarding assessment (see appendix 1) showed that there are not principles with a need to be monitored.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project reflects the key gender issues and requirements of gender-sensitive design and implementation of the project. SDG#5 is one of the impact areas of the project. The introduction of new and clean cookstoves will result in reducing use of charcoal at the user's place. In Uganda, cooking activity at household level is mostly still managed by women. Therefore, women are more exposed to the indoor air pollution and the associated hazard and will benefit more from clean cookstoves. Time savings due to faster cooking also provides special benefits to women.

Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The Uganda National Gender Policy from 2007 defines principles towards a society free from gender-based discrimination in many different areas of daily life[23]. The project aligns with this policy by providing practical benefits to women who use the clean stoves (see above), but also by directly dealing with women as main and fully responsible customers and workers, thereby fully respecting women's equal rights and economic independence.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	All the gender safeguarding principles and requirements are carefully assessed and an expert is not required.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	The project team responsible for organizing the stakeholder consultation did ensure the gender representation at the meeting. So, no expert is required.

[23] https://evaw-global-database.unwomen.org/en/countries/africa/uganda/2007/national-gender-policy--2007-

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

The physical LSC meeting tool place on May 20 2022 in Mbale, with over 70 participants. The project and planned GS certification procedures were explained and discussed, including approval for the transfer of owner ship on carbon credits. Also, basic issues for the inclusion of further VPAs like this one were discussed, as well as probable impacts and SDG outcomes of the VPA.
No mitigation measures have been identified during the consultation.

E.2. Final continuous input / grievance mechanism

METHOD	INCLUDE ALL DETAILS OF CHOSEN METHOD (S) SO THAT THEY MAY BE UNDERSTOOD AND, WHERE RELEVANT, USED BY READERS.
Continuous Input / Grievance Expression Process Book (mandatory)	ACE Office, Senior Quarters, Mbale (Next to Crown Suites Hotel)
GS Contact (mandatory)	help@goldstandard.org
Telephone access (optional)	+256 755368077
Internet/email access (optional)	tyler@africancleanenergy.com
The Nominated Independent Mediator (optional)	N/A, Our Stakeholders were not interested in this option

SECTION F. Eligibility and inclusion criteria for VPAs inclusion

The below table shall be completed for all VPAs.

The CME shall provide clear description on how eligibility criteria set at real case VPAs are complied with for each real case and regular VPAs submitted for inclusion.

The CME shall not change the eligibility criteria and required condition set at real case VPAs. At the time of inclusion of regular VPAs, the CME shall only describe how the regular VPAs comply with the eligibility criterion.

NO.	ELIGIBILITY CRITERION	DESCRIPTION/ REQUIRED CONDITION	DESCRIPTION OF THE VPA IN RELATION TO THE CRITERIA, MEANS OF VERIFICATION/SUPPORTING EVIDENCE FOR INCLUSION
1	Technology	ACE One stoves are used	ACE One stoves are used (see A.3)
2	Project area	The VPA is implemented in Uganda	Implemented in Uganda (See A.2)
3	VPA implementer	The VPA is implemented by ACE Uganda	Implemented by ACE Uganda (see KPI)
4	PoA eligibility	The VPA complies with all PoA eligibility criteria	Compliance shown in A.1.1

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into SECTION D above. Please refer to the instructions in the Guide to Completing this Form below.

SOCIAL SAFEGUARDING PRINCIPLES		
Reference requirement	Question	Response
P.1 |HUMAN RIGHTS		
P.1.1.1 |	Does the project developer, its representatives and the Project disrespect internationally proclaimed human rights?	☐ YES ☒ NO
P.1.1.1 |	Is the project involved or complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights?	☐ YES ☒ NO
P.1.1.2 |	Have local communities or individuals raised human rights concerns regarding the project (e.g., during the stakeholder engagement process, grievance processes, public statements)?	☐ YES ☒ NO
P.1.1.3 |	Is there a risk that rights-holders (e.g., Project-affected stakeholders) do not have the capacity to claim their rights?	☐ YES ☒ NO
P.1.1.3 |	Does this project undermine national or regional measures for the realisation of the right to development?	☐ YES ☒ NO
If the answer to any of the questions above is "yes," please explain the reason and how the project will ensure compliance with applicable requirements.		
Please add text here…		
Would the project potentially involve or lead to:		
P.1.1.1 |	adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalised groups?	☐ YES ☐ POTENTIALLY ☒ NO
P.1.1.2 |	inequitable or discriminatory impacts on affected populations, particularly people living in poverty or marginalised or excluded individuals or groups, including persons with disabilities?	☐ YES ☐ POTENTIALLY ☒ NO

P.1.1.3 |	restrictions in availability, quality of and/or access to resources or basic services, in particular to marginalised individuals or groups, including persons with disabilities?	☐ YES ☐ POTENTIALLY ☒ NO
P.1.1.3 |	exacerbation of conflicts among and/or the risk of violence to project-affected communities and individuals?	☐ YES ☐ POTENTIALLY ☒ NO

Briefly describe below how the project incorporates a human rights-based approach.
For example, by describing how the project design:

- is informed by human rights analysis, including from UN human rights mechanisms (human rights treaty bodies, universal periodic review, special procedures)
- includes measures to assist the government to realise (respect, protect and fulfil) human rights under international law and to implement human rights-related standards in national law (whichever is higher)
- enhances the availability, accessibility and quality of benefits and services for potentially marginalised individuals and groups, and to increase their inclusion in decision-making processes that may impact them (consistent with the non-discrimination and equality human rights principle)
- provides reasonable accommodations to strengthen inclusivity and accessibility of project benefits and services to persons with disabilities.

1. The project does not negatively affect human rights. Participation is fully voluntary.

2. No discrimination occurs because the project offers clean cookstoves to interested people who can freely decide to use them or not. The project is basically targeting women in poor households depending on charcoal for cooking

P.2 | GENDER EQUALITY AND WOMEN'S EMPOWERMENT

P.2.1.1 |	Have women's groups/leaders raised gender equality concerns regarding the project, (e.g., during the stakeholder engagement process, grievance processes, public statements)?	☐ YES ☒ NO
P.2.1.2 |	Does the project undermine the principles of non-discrimination, equal treatment, and equal pay for equal work?	☐ YES ☒ NO
P.2.1.2 |	Does the project prevent men and women from having equal opportunities to participate in identified tasks and activities, whether through paid work, volunteer work, or community contributions, as appropriate?	☐ YES ☒ NO
P.2.1.2 |	Does the project limit the participation of women or men based on pregnancy, maternity/paternity leave, or marital status?	☐ YES ☒ NO
P.2.1.2 |	Is information about project objectives being communicated in a way that is inappropriate for the local context and not tailored to the methods of understanding of both women and men, which could hinder their participation?	☐ YES ☒ NO

P.2.1.3 |	Has the project assessed gender risks without referencing the country's gender strategy or equivalent national commitment?	☐ YES ☒ NO
P.2.1.4 |	Has expert stakeholder(s) been involved, and has their input been requested for the project design on gender equality and women's empowerment?	☐ YES ☒ NO
If the answer to any of the questions above is "yes," please explain the reason and how the project will ensure compliance with applicable requirements.		
Please add text here….		
Would the project potentially involve or lead to:		
P.2.1.1 |	adverse impacts on gender equality and/or the situation of women and girls?	☐ YES ☐ POTENTIALLY ☒ NO
P.2.1.1 |	exacerbation of risks of gender-based violence? For example, through the influx of workers to a community, changes in community and household power dynamics, increased exposure to unsafe public places and/or transport, etc.	☐ YES ☐ POTENTIALLY ☒ NO
P.2.1.2 |	reproducing discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	☐ YES ☐ POTENTIALLY ☒ NO
P.2.1.2 |	limitations on women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well-being.	☐ YES ☐ POTENTIALLY ☒ NO
Briefly describe below how the project is addressing any identified risk to gender equality and women's empowerment.		
1. The project focuses on a gender sensitive design and planning. Women were involved from the beginning when exploring the feasibility of the project and designing its implementation. Furthermore, the Stakeholder Consultation was organized with a majority of participants being women. 2. The project does apply the principles of non-discrimination and equal treatment in distribution and usage of the products. It embraces the spirit of the Uganda National Gender Policy[24] and the national		

[24] https://evaw-global-database.unwomen.org/en/countries/africa/uganda/2007/national-gender-policy--2007-

Labour Laws[25] and ILO guidelines. ACE also ensures equal pay for equal work during any activity in the project.

3. The project applies the principles of Uganda National Gender Policy (see section D.2).

P.3 |COMMUNITY HEALTH AND SAFETY

P.3.1.1 |	Does the project involve potential risks to the health and safety of affected communities during its life cycle?	☐ YES ☒ NO
P.3.1.2 |	Does the project involve any potential risks to the workers' safety and health?	☐ YES ☒ NO

If the answer to any of the questions above is "yes," please explain the reason and how the project will ensure compliance with applicable requirements.

Please add text here….

Would the project potentially involve or lead to:

P.3.1.1 |	construction and/or infrastructure development (e.g., roads, buildings, dams)?	☐ YES ☒ NO
P.3.1.2 |	air pollution, noise, vibration, traffic, injuries, physical hazards, poor surface water quality due to runoff, erosion, sanitation?	☐ YES ☐ POTENTIALLY ☒ NO
P.3.1.2 |	harm or losses due to failure of structural elements of the project (e.g., collapse of buildings or infrastructure)?	☐ YES ☐ POTENTIALLY ☒ NO
P.3.1.2 |	risks of water-borne or other vector-borne diseases (e.g., temporary breeding habitats), communicable and noncommunicable diseases, nutritional disorders, mental health?	☐ YES ☐ POTENTIALLY ☒ NO
P.3.1.2 |	transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g., explosives, fuel and other chemicals during construction and operation)?	☐ YES ☐ POTENTIALLY ☒ NO
P.3.1.2 |	adverse impacts on ecosystems and ecosystem services relevant to communities' health (e.g., food, surface water purification, natural buffers from flooding)?	☐ YES ☐ POTENTIALLY ☒ NO

Briefly describe below how the project is addressing any identified risk related to community health and safety.

[25] https://mglsd.go.ug/laws/

There are no health risks associated to the usage of the improved cookstoves offered. To the contrary, the reduction of kitchen smoke is an important objective of the programme and part of monitoring under SDG 3. Due to the principle of pyrolysis producing wood gas, the stoves burn as clear as LPG stoves.

P.4 |CULTURAL HERITAGE, INDIGENOUS PEOPLE, DISPLACEMENT AND RESETTLEMENT

P.4.1 |Sites of Cultural and Historical Heritage

P.4.1.1 |	Does the project involve altering, damaging, or removing sites, objects, or structures of significant cultural heritage?	☐ YES ☒ NO

If the answer to question above is "yes," please explain the reason and how the project will ensure compliance with applicable requirements.

Please add text here….

Would the project potentially involve or lead to:

P.4.1.1 |	activities adjacent to or within a cultural heritage site?	☐ YES ☐ POTENTIALLY ☒ NO
P.4.1.1 |	significant excavations, demolitions, movement of earth, flooding or other environmental changes?	☐ YES ☐ POTENTIALLY ☒ NO
P.4.1.1 |	alterations to landscapes and natural features with cultural significance?	☐ YES ☐ POTENTIALLY ☒ NO
P.4.1.1 |	adverse impacts to sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g., knowledge, innovations, practices)? (Note: projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	☐ YES ☐ POTENTIALLY ☒ NO
P.4.1.2 |	utilisation of tangible and/or intangible forms (e.g., practices, traditional knowledge) of Cultural Heritage for commercial or other purposes?	☐ YES ☐ POTENTIALLY ☒ NO
P.4.1.2 |	If answer to question above is “YES” or “POTENTIALLY” - are the communities made aware of their right under the law, scope and nature of proposed development and its potential consequences?	☐ YES ☐ NO ☒ NA
P.4.1.3 |	If answer to question above is “YES” - does the project provide equitable sharing of benefits from commercialisation of such knowledge, innovation, or practice, consistent with their customs and traditions?	☐ YES ☐ NO ☒ NA

P.4.1.4 |	If answer to question above is "YES" - are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA
P.4.1.4 |	If answer to question above is "YES", has project design been changed, modified, updated considering opinions and recommendations of an Expert Stakeholder?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

Please add text here….

P.4.2 |FORCED EVICTION AND DISPLACEMENT

P.4.2.1 |	Does the project involve any risks related to involuntary relocation of people?	☐ YES ☒ NO

If the answer to question above is "yes," please explain the reason and how the project will ensure compliance with applicable requirements.

Please add text here….

Would the project potentially involve or lead to:

P.4.2.1 |	risk of forced evictions or involuntary relocation of people?	☐ YES ☐ POTENTIALLY ☒ NO
P.4.2.2 |	temporary or permanent and full or partial physical displacement (including people without legally recognisable claims to land)?	☐ YES ☐ POTENTIALLY ☒ NO
P.4.2.2 |	economic displacement (e.g., loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	☐ YES ☐ POTENTIALLY ☒ NO
P.4.2.2 |	If answer to question above is "YES" or "POTENTIALLY", - has the project developed Resettlement Action Plan or Livelihood Action Plan in consultation and agreement with affected individual, group or community? - has the project integrated Resettlement Action Plan or Livelihood Action Plan into the Project design?	☐ YES ☐ NO ☒ NA
P.4.2.3 |	If answer to question above is "YES" - are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA
P.4.2.3 |	If answer to question above is "YES", have project design been changed, modified, updated considering opinions and recommendations of an Expert Stakeholder?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.
Please add text here….

P.4.3 |LAND TENURE AND OTHER RIGHTS

P.4.3.1 |	Does the project involve any risks related to identifying and managing legitimate tenure rights that may be affected by the project?	☐ YES ☒ NO
If the answer to question above is "yes," please explain the reason and how the project will ensure compliance with applicable requirements.		
Please add text here….		
Would the project potentially involve or lead to:		
P.4.3.1 |	impacts on or changes to land tenure arrangements and/or community-based property rights/customary rights to land, territories and/or resources?	☐ YES ☐ POTENTIALLY ☒ NO
P.4.3.1 |	uncertainties with regards to land tenure, access rights, usage rights or land ownership? Examples include, but are not limited to water access rights, community-based property rights and customary rights.	☐ YES ☐ POTENTIALLY ☒ NO
P.4.3.2 |	Changes in legal arrangements, if yes, are the changes done in line with relevant laws and regulations?	☐ YES ☐ NO ☒ NA
P.4.3.2 |	Changes in legal arrangements, if yes, are these changes agree with free, prior and informed consent of the involved stakeholders?	☐ YES ☐ NO ☒ NA
P.4.3.3 |	Does some other entity (other than the project developer) hold uncontested land title for the entire Project Boundary?	☐ YES ☐ NO ☒ NA
P.4.3.4 |	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA
P.4.3.4 |	If answer to question above is "YES", have project design been changed, modified, updated considering opinions and recommendations of an Expert Stakeholder?	☐ YES ☐ NO ☒ NA
P.4.3.5 |	Have project developer in consultation with stakeholders established a functioning mechanism to receive, process, resolve, communicate and record grievances?	☒ YES ☐ NO ☐ NA
If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.		

Please add text here….

P.4.4 |INDIGENOUS PEOPLES

P.4.4.1 |	Does the project involve Indigenous People within the Project area of influence who may be affected directly or indirectly by the Project?	☐ YES ☒ NO
If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.		
Please add text here….		
Would the project potentially involve or lead to:		
P.4.4.1 |	affect areas where indigenous peoples are present (including project area of influence)	☐ YES ☐ POTENTIALLY ☒ NO
P.4.4.1 |	affect areas, land and territory claimed by indigenous peoples?	☐ YES ☐ POTENTIALLY ☒ NO
P.4.4.1 |	impacts (positive or negative) to the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples?	☐ YES ☐ POTENTIALLY ☐ NO
P.4.4.7 |	If answer to above questions is ''YES'' or "POTENTIALLY", - Is it determined that the proposed project may affect the rights, lands, resources, or territories of indigenous people? - Has an "Indigenous People Plan" (IPP) or "Indigenous People Plan Framework" been elaborated and included in the project documentation? - Was the plan developed in accordance with the effective and meaningful participation of indigenous peoples and in accordance with UNDP Guidelines?	☐ YES ☐ NO ☒ NA
P.4.4.3 |	risk of forcibly removing indigenous people from their lands and territories?	☐ YES ☐ POTENTIALLY ☒ NO
P.4.4.4 |	utilisation and/or commercial development of natural resources on lands and territories claimed by indigenous peoples? Consider, and where appropriate ensure, consistency with the answers under Principle 4.1 above	☐ YES ☐ POTENTIALLY ☒ NO
P.4.4.5 |	If answer to question above is "YES" or "POTENTIALLY"	☐ YES

P.4.4.6 |	- Did the project obtain free, prior and informed consent from indigenous people before taking their cultural, intellectual, religious, and/or spiritual property? - Does the project ensure that the indigenous people receive an equitable sharing of benefits resulting from the use of their traditional knowledge and practices? ? - Does the project ensure that the sharing of benefits resulting from the use of indigenous peoples' traditional knowledge and practices is culturally appropriate and inclusive? - Does the project ensure that the provision of equitable sharing of benefits does not impede land rights or equal access to basic services including health services, clean water, energy, education, safe and decent working conditions, and housing?	☐ NO ☒ NA
P.4.4.8 |	Does the project lack appropriate feedback and grievance channels for Indigenous Peoples and their representatives?	☐ YES ☐ NO ☒ NA
P.4.4.8 |	Has a grievance mechanism not been established at the beginning of programme or project implementation with due consideration given to customary dispute settlement mechanisms among the Indigenous Peoples concerned and will it remain operational throughout the project cycle?	☐ YES ☐ NO ☒ NA
P.4.4.9 |	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA
P.4.4.9 |	If answer to question above is “YES”, have project design been changed, modified, updated considering opinions and recommendations of an Expert Stakeholder?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

Please add text here….

P.5 |CORRUPTION

P.5.1.1 |	Does the project involve, or is it complicit in, contributing to or reinforcing corruption or corrupt projects?	☐ YES ☒ NO
P.5.1.1 |	Does the project have a risk of encouraging bribery, kickbacks, or other unethical behavior?	☐ YES ☒ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.		
Please add text here....		
ECONOMIC SAFEGUARDING PRINCIPLES		
P.6 |ECONOMIC IMPACTS		
P.6.1 |LABOUR RIGHTS AND WORKING CONDITIONS		
P.6.1.1 |	Does the project involve, facilitate, or condone forced labor, or pose a potential risk of forced labor?	☐ YES ☒ NO
P.6.1.1 |	Does the project violate any labor or health and safety laws, international obligations, or ILO conventions?	☐ YES ☒ NO
P.6.1.2 |	Does the project violate the principles of equal opportunity and fair treatment in its employment decisions?	☐ YES ☒ NO
P.6.1.3 |	Does the project violate national laws, if available regarding non-discrimination in employment?	☐ YES ☒ NO
P.6.1.4 | P.6.1.5 |	Does the project allow child labor?	☐ YES ☒ NO
P.6.1.7 | P.6.1.8 |	Does the project have insufficient processes and measures in place to ensure the safety and health of project workers?	☐ YES ☒ NO
P.6.1.9 |	Does the project have insufficient measures to safeguard and support vulnerable project workers, such as women, people with disabilities, migrant workers, and young workers, and to prevent any kind of harassment, abuse, bullying, or exploitation, including gender-based violence (GBV)?	☐ YES ☒ NO
P.6.1.10 |	Does the project have no grievance mechanism available for workers to voice workplace concerns? Is information about this mechanism not provided to workers at the time of recruitment, or is it not easily accessible?	☐ YES ☒ NO
If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.		
Please add text here....		
Would the project potentially involve or lead to: (NOTE: APPLIES TO BOTH PROJECT AND CONTRACTOR WORKERS)		
P.6.1.1 |	use of forced labour?	☐ YES ☐ POTENTIALLY ☒ NO
P.6.1.1 |	working conditions that do not meet national labour laws and international commitments?	☐ YES ☐ POTENTIALLY

		☒ NO
P.6.1.1 |	working conditions that may deny freedom of association and collective bargaining?	☐ YES ☐ POTENTIALLY ☒ NO
P.6.1.1 |	absence of documented working agreements with all individual workers *if such agreements do not exist, or do not address working conditions and terms of employment, the project developer shall provide reasonable working conditions and terms of employment.*	☐ YES ☐ POTENTIALLY ☒ NO
P.6.1.1 |	use of migrant workers? *if engaged, the developer shall ensure that they are engaged substantially equivalent terms and conditions to non-migrant workers carrying out similar work.*	☐ YES ☐ POTENTIALLY ☒ NO
P.6.1.1 |	having no arrangements for basic services[26] for workers? *the project developer shall put in place and implement policies on the quality and management of the accommodation and provision of basic services in a manner consistent with the principles of non-discrimination and equal opportunity. Workers' accommodation arrangements should not restrict workers' freedom of movement or of association*	☐ YES ☐ POTENTIALLY ☒ NO
P.6.1.2 |	any form of discrimination or harassment based on factors unrelated to job requirements, such as gender, race, nationality, ethnicity, social or indigenous origin, religion or belief, disability, age, or sexual orientation?	☐ YES ☐ POTENTIALLY ☒ NO
P.6.1.2 |	any form of discrimination in any aspect of employment, such as recruitment, compensation, working conditions, training, job assignment, promotion, termination, or discipline?	☐ YES ☐ POTENTIALLY ☒ NO
P.6.1.2 |	harassment, intimidation, and/or exploitation, especially in regard to women?	☐ YES ☐ POTENTIALLY ☒ NO
P.6.1.3 |	discriminatory working conditions and/or lack of equal opportunity where national law provides provision to address non-discrimination in employment?	☐ YES ☐ POTENTIALLY ☒ NO

[26] Basic services requirements refer to minimum space, supply of water, adequate sewage and garbage disposal system, appropriate protection against heat, cold, damp, noise, fire, and disease-carrying animals, adequate sanitary and washing facilities, ventilation, cooking and storage facilities and natural and artificial lighting, and in some cases basic medical services.

P.6.1.4 |	use of child labour? (including third-party engaged workers)	☐ YES ☐ POTENTIALLY ☒ NO
P.6.1.4 |	inadequate and verifiable mechanisms for age verification?	☐ YES ☒ NO
P.6.1.7 |	no processes and measures in place for the safety and health of project workers?	☐ YES ☒ NO
P.6.1.7 |	No provision of safety and health training provisions, including on the proper use and maintenance of personal protective equipment conducted by competent persons and the maintenance of training records?	☐ YES ☒ NO
P.6.1.7 |	No provision to record and document accidents, diseases, incidents, and any resulting injuries, illnesses, or deaths?	☐ YES ☒ NO
P.6.1.8 |	occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle?	☐ YES ☒ NO
P.6.1.9 |	No measures to protect vulnerable project workers from harassment, exploitation, and gender-based violence (GBV)? This includes women, people with disabilities, migrant workers, and young workers.	☐ YES ☒ NO
P.6.1.10 |	No grievance mechanism available for workers to voice workplace concerns.	☐ YES ☒ NO
P.6.1.11 |	No measures for due diligence and the establishment of policies and procedures to manage and monitor the performance of third-party employees in the project?	☐ YES ☒ NO

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

Please add text here….

P.6.2 |NEGATIVE ECONOMIC CONSEQUENCES

P.6.2.1 |	Is there a risk of project failure during implementation or after project certification due to a lack of financial resources?	☐ YES ☒ NO
P.6.2.2 |	Does the project have potential negative impacts or pose a risk to the local economy?	☐ YES ☒ NO
P.6.2.2 |	Are there any potential risks or negative impacts this project may have on vulnerable or marginalised social groups, despite the benefits it may bring?	☐ YES ☒ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

Please add text here…

Would the project involve or lead to:		
P.6.2.2 |	economic impacts (negative/detrimental) to the local economy?	☐ YES ☐ POTENTIALLY ☒ NO
P.6.2.2 |	negative economic consequences during and after project implementation, e.g., for vulnerable and marginalised social groups in targeted communities?	☐ YES ☐ POTENTIALLY ☒ NO
If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.		
Please add text here….		
P.7 |CLIMATE AND ENERGY		
P.7.1 |GHG EMISSIONS		
P.7.1.1 |	Does the project have a risk of increasing greenhouse gas emissions over the Baseline Scenario?	☐ YES ☒ NO
If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.		
Please add text here….		
Would the project involve or lead to:		
P.7.1.1 |	increase greenhouse gas emissions over the Baseline Scenario?	☐ YES ☐ POTENTIALLY ☒ NO
If the answer is "yes" or "potentially" to the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.		
Please add text here….		
P.7.2 |ENERGY SUPPLY		
P.7.2.1 |	Does the project pose a risk to the availability and reliability of energy supply to other users?	☐ YES ☒ NO
If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.		
Please add text here….		

Would the project involve or lead to:		
P.7.2.1 |	negative impact on the availability and reliability of energy supply to other users?	☐ YES ☐ POTENTIALLY ☒ NO

If the answer is "yes" or "potentially" to the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

Please add text here….

P.8 |WATER

P.8.1 |IMPACT ON NATURAL WATER PATTERNS/FLOWS

P.8.1.1 |	Does the project increase water usage to a level that will not allow for the maintenance of environmental flows?	☐ YES ☒ NO
P.8.1.1 |	Does the project result in the discharge of wastewater that does not meet the required standard for beneficial reuse and could therefore negatively impact the environmental flow?	☐ YES ☒ NO
P.8.1.1 |	Does the project have the potential risk to exceed the rate of recharge for the groundwater source?	☐ YES ☒ NO
P.8.1.1 |	Does the project involve any processes or activities that could contaminate the groundwater and render it unsuitable for use?	☐ YES ☒ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

Please add text here….

Would the project involve or lead to:		
P.8.1.1 |	affect the natural or pre-existing pattern of watercourses, groundwater and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	☐ YES ☐ POTENTIALLY ☒ NO
P.8.1.1 |	Wastewater discharge of quality that does not meet the required standard for beneficial reuse?	☐ YES ☐ POTENTIALLY ☒ NO
P.8.1.1 |	significant extraction, diversion of ground water? For example, construction of dams, reservoirs, river basin developments, groundwater extraction	☐ YES ☐ POTENTIALLY ☒ NO
P.8.1.2 |	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.
Please add text here….

P.8.2 |EROSION AND/OR WATER BODY INSTABILITY

P.8.2.1 |	Does the project have a risk of negatively impacting the catchment and has it been assessed and addressed?	☐ YES ☒ NO

If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.
Please add text here….
Would the project involve or lead to:

P.8.2.2 | - P.8.2.5 |	negatively impact on the catchment area? *If yes, Erosion prevention measures, including soil and slope protection measures, must be implemented before project commencement. These measures should involve natural terracing, infiltration strips, permanent ground cover, hedge and tree rows, and effective slope length assessment. Regular reassessment of these measures is necessary.*	☐ YES ☐ POTENTIALLY ☒ NO
P.8.2.6 |	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.
Please add text here….

P.9 |ENVIRONMENT, ECOLOGY AND LAND USE

P.9.1 |LANDSCAPE MODIFICATION AND SOIL

P.9.1.1 | - P.9.1.3 |	Is there any risk of soil resource degradation or loss of ecosystem services provided by soils in the project? *If yes, the project shall maintain healthy soils by minimising negative impacts on soil health, productivity, structure, and water retention. Steps to minimise soil degradation include crop rotation, composting, using N-*	☐ YES ☒ NO

	fixing plants, and reducing tillage and ecologically harmful substances.	
If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.		
Please add text here….		
Would the project involve or lead to:		
P.9.1.4 |	production, harvesting, and/or management of living natural resources by small-scale landholders and/or local communities?	☐ YES ☐ POTENTIALLY ☒ NO
P.9.1.4 |	if answer to above question "yes" or "potentially", does project adopt appropriate and culturally sensitive sustainable resource management practices?	☐ YES ☐ NO ☒ NA
If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.		
Please add text here….		
P.9.2 |VULNERABILITY TO NATURAL DISASTER		
P.9.2.1 |	Does the project have any risks associated with natural or man-made hazards that could result from land use changes due to the project?	☐ YES ☒ NO
If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.		
Please add text here….		
Would the project involve or lead to:		
P.9.2.2 |	any potential risks that require emergency preparedness and response planning?	☐ YES ☐ POTENTIALLY ☒ NO
P.9.2.2 |	if answer to above question "yes" or "potentially", did the project developer disclose appropriate information about emergency preparedness and response to affected communities?	☐ YES ☐ NO ☒ NA
If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.		
Please add text here….		
P.9.3 |BIOSAFETY AND GENETIC RESOURCES		

P.9.3.1 |	Does the project involve the transfer, handling, and use of genetically modified organisms/living modified organisms that may result in adverse effects on biological diversity?	☐ YES ☒ NO

If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

Please add text here….

Would the project involve or lead to:

P.9.3.1 |	the transfer, handling and use of genetically modified organisms/living modified organisms (GMOs/LMOs) that result from modern biotechnology	☐ YES ☐ POTENTIALLY ☒ NO
P.9.3.1 |	If answer to above question is "yes" has a risk assessment by a competent Expert stakeholder been carried out in accordance with Annex iii of the Cartagena protocol on biosafety to the convention on biological diversity?	☐ YES ☒ NO ☐ NA
P.9.3.2 |	If answer to above question is "yes" has any risks identified in the risk assessment?	☐ YES ☐ NO ☒ NA
P.9.3.3 |	Forestry (for example Afforestation/Reforestation) involving GMO planting? *Note - Forestry projects (for example Afforestation/ Reforestation) involving GMO planting are not eligible for Certification under Gold Standard for the Global Goals.*	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

Please add text here….

P.9.4 |RELEASE OF POLLUTANTS

P.9.4.1 |	Does the project have a risk of releasing pollutants to air, water, and land in routine, non-routine, or accidental circumstances?	☐ YES ☒ NO

If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

Please add text here….

Would the project involve or lead to:

P.9.4.1 |	any potential risk of pollutant release that cannot be avoided?	☐ YES ☐ POTENTIALLY ☒ NO

P.9.4.3 |	If answer to above question is "Yes" or "potentially", has the project identified all potential pollution sources that may degrade the quality of soil, air, surface, and groundwater in the project area?	☐ YES ☐ NO ☒ NA
P.9.4.2 |	If answer to above question is "Yes" or "potentially", do the pollution prevention and control technologies and practices applied during the project life cycle align with national regulations or international best practices?	☐ YES ☐ NO ☒ NA
P.9.4.3 |	If answer to above question is "Yes", is there a monitoring plan to ensure that mitigation measures are implemented, and resources are protected?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

Please add text here….

P.9.5 |HAZARDOUS AND NON-HAZARDOUS WASTE

P.9.5.1 |	Does the project involve the generation of waste materials (both hazardous and non-hazardous)?	☐ YES ☒ NO
P.9.5.3 |	Does the project involve risk of release of hazardous materials resulting from their production, transportation, handling, storage, or use?	☐ YES ☒ NO
P.9.5.5 |	Does the project involve the use of any chemicals or materials subject to international bans or phase-outs?	☐ YES ☒ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

Please add text here….

Would the project involve or lead to:

P.9.5.1 |	the generation and management of waste materials?	☐ YES ☐ POTENTIALLY ☒ NO
P.9.5.1 |	treatment, destruction, or disposal of waste material?	☐ YES ☐ NO ☒ NA
P.9.5.1 |	If answer to above question is "Yes", does the project involve an environmentally friendly method that includes appropriate control of emissions and residues resulting from the handling and processing of waste material?	☐ YES ☐ NO ☒ NA
P.9.5.3 |	risk of release of hazardous materials resulting from their production, transportation, handling, storage, or use?	☐ YES ☒ NO ☐ NA

P.9.5.3 |	If answer to above question is "yes", does project has measures in place to address health risks?	☐ YES ☐ NO ☒ NA
P.9.5.4 |	Involve manufacture, trade, and use of chemicals and hazardous materials subject to international bans or phase-outs due to their high toxicity to living organisms, environmental persistence, potential for bioaccumulation, or potential for depletion of the ozone layer	☐ YES ☐ POTENTIALLY ☒ NO

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

Please add text here….

P.9.6 |PESTICIDES & FERTILISERS

P.9.6.1 |	Does the project involve the use of chemical pesticides?	☐ YES ☒ NO
P.9.6.5 |	Does the project involve purchase, store, manufacture, trade or use products that fall in Classes IA (extremely hazardous) and IB (highly hazardous)	☐ YES ☒ NO
P.9.6.6 |	Does the project use fertilisers, and if so, are measures being taken to minimise their use and nutrient losses to the environment?	☐ YES ☒ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

Please add text here….

Would the project involve or lead to:

P.9.6.1 |	chemical pesticides use for pest management?	☐ YES ☐ POTENTIALLY ☒ NO
P.9.6.4 |	If answer to question above is "yes" or "potentially", does project has documented Chemical Pesticides Policy in place?	☐ YES ☐ NO ☒ NA
P.9.6.5 |	purchase, store, use, manufacture, or trade in Class II (moderately hazardous) pesticides?	☐ YES ☐ POTENTIALLY ☒ NO
P.9.6.5 |	If answer to question above is "yes" or "potentially", does project has appropriate controls on manufacture, procurement, or distribution and/or use of these chemicals?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above questions, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

Please add text here….

P.9.7 |HARVESTING OF FORESTS

P.9.7.1 |	Does the project have a risk of unsustainable forest management, including timber harvesting?	☐ YES ☒ NO
P.9.7.1 |	Does the project pose a risk of depleting biodiversity and ecosystem functionality in areas where improved forest management is undertaken?	☐ YES ☒ NO
P.9.7.1 |	Does the project risk not meeting requirements for environment-friendly, socially beneficial, and economically viable plantations using native species whenever possible?	☐ YES ☒ NO

If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

Please add text here….

P.9.8 |FOOD SECURITY

P.9.8.1 |	Does the project involve the risk of negatively influencing access to and availability of food for people affected?	☐ YES ☒ NO

If the answer to the question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

Please add text here….

Would the project involve or lead to:

P.9.8.1 |	modification of the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	☐ YES ☐ POTENTIALLY ☒ NO

If the answer is "yes" or "potentially" to the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

Please add text here….

P.9.9 | ANIMAL WELFARE

P.9.9.1 |	Does the project involve any risks to animal welfare? Animal welfare shall be ensured by providing access to water and food, appropriate environment, humane treatment, and staff training. Evidence of mistreatment will be treated as an immediate non-conformity.	☐ YES ☒ NO
P.9.9.2 |	Does the project involve any potential risk of excessive or inadequate use of veterinary medicines?	☐ YES ☒ NO
P.9.9.4 |	Does the project involve the risk of administering synthetic growth promoters, including hormones?	☐ YES

		☒ NO
If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.		
Please add text here….		
Would the project involve or lead to:		
P.9.9.1 |	animal husbandry or harvesting of fish populations or other aquatic species?[27]	☐ YES ☐ NO ☒ NA
P.9.9.1 |	limiting access for animals to basic needs like drinking water, adequate food, daylight, appropriate shelter etc.?	☐ YES ☐ POTENTIALLY ☒ NO
P.9.9.3 |	inadequate measures to isolate sick animals and control the spread of disease, especially zoonotic diseases?	☐ YES ☐ NO ☒ NA
P.9.9.5 |	inadequate low-stress methods, equipment, and facilities that facilitate calm animal movement.	☐ YES ☐ NO ☒ NA
P.9.9.6 |	inadequate measures to ensure that animals are exposed to the least stress possible during transportation and slaughtering?	☐ YES ☐ NO ☒ NA
P.9.9.7 |	inappropriate spacing per animal and stocking rates per land unit?	☐ YES ☐ NO ☒ NA
P.9.9.8 |	inadequate measures to address the specific needs of aquatic animals?	☐ YES ☐ NO ☒ NA
P.9.9.9 | P.9.9.10 |	primary production of living natural resources such as animal husbandry, aquaculture, and fisheries? If the answer is yes, implement industry-standard sustainable management practices in line with to one or more relevant and credible standards and utilise available technologies.	☐ YES ☐ NO ☒ NA

[27] 'Involve' means if the project mechanism and/or impact(s) are achieved via changing animal husbandry practices in some way.

If the answer is "yes" or "potentially" to any of the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.
Please add text here….

P.9.10 |HIGH CONSERVATION VALUE AREAS AND CRITICAL HABITATS

P.9.10.1 |	Does the project have the risk of negatively impacting HCV areas and/or critical habitats?	☐ YES ☒ NO
P.9.10.2 |	Does the project in the project area or area of downstream impacts have risks to the following: native tree patches, individual native trees, freshwater resources (including rivers, lakes, swamps, temporary water bodies, and wells), habitats of rare, threatened, and endangered species, and biodiversity-enhancing areas?	☐ YES ☒ NO
If the answer to any of the questions above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.		
Please add text here….		
Would the project involve or lead to:		
P.9.10.1 |	identified habitats as HCV areas and or Critical habitats?	☐ YES ☐ POTENTIALLY ☒ NO
P.9.10.1 |	If answer to above question is "yes", does the project have any risks that could negatively impact the catchment, project success, and surrounding HCV and ecological assets, as well as any measurable adverse impacts on the criteria or biodiversity values for which the critical habitat was designated, and on the ecological processes supporting that biodiversity?	☐ YES ☐ NO ☒ NA
P.9.10.1 |	If answer to above question is "yes", is a robust, appropriately designed, and long-term Habitats and Biodiversity Action Plan absent which will make the project unable to achieve net gains of those biodiversity values for which the critical habitat was designated?	☐ YES ☐ NO ☒ N/A
P.9.10.2 |	Does the project area or area of downstream impacts have native tree patches, individual native trees, freshwater resources (including rivers, lakes, swamps, temporary water bodies, and wells), habitats of rare, threatened, and endangered species, and biodiversity-enhancing areas?	☐ YES ☐ POTENTIALLY ☒ NO
P.9.10.2 |	If the answer to the above question is "yes", will the project have any adverse effects on these areas?	☐ YES ☐ No ☒ NA
P.9.10.3 |	If the answer to above question is "yes", does the project has opportunities to minimise unwarranted conversion or	☐ YES ☐ No

	degradation of the habitat and to enhance the habitat as part of its development?	☒ NA
P.9.10.4 |	Is the project applying Land Use & Forest Activity Requirements and managing a minimum 10% of the project area to protect or enhance the biological diversity of native ecosystems following HCV approach as per the given requirements?	☐ YES ☐ No ☒ NA
P.9.10.5 |	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

Please add text here….

P.9.11 |ENDANGERED SPECIES

P.9.11.1 |	Does the project lead to the reduction or negative impact on any recognised Endangered, Vulnerable or Critically Endangered species?	☐ YES ☒ NO

If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.

Please add text here….

Would the project involve or lead to:

P.9.11.2 |	distortion of habitats of endangered species?	☐ YES ☐ POTENTIALLY ☒ NA
P.9.11.2 |	If answer to the above question is “yes”, does the project plan to protect and enhance them?	☐ YES ☐ POTENTIALLY ☐ NO ☒ N/A
P.9.11.2 |	Are opinions and recommendations of an Expert Stakeholder(s) not sought and demonstrated as being included in the project design?	☐ YES ☐ NO ☒ NA

If the answer is "yes" or "potentially" to any of the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.

Please add text here….

P.9.12 |INVASIVE ALIEN SPECIES

P.9.12.1 |	Does project introduce any alien species (not currently established in the country or region of the project) into new environments?	☐ YES ☒ NO
If the answer to question above is "yes," please explain project situation and how the project will ensure compliance with applicable requirements.		
Please add text here….		
Would the project involve or lead to:		
P.9.12.1 |	risk of introducing any alien species with a high risk of invasive behaviour regardless of whether such introductions are permitted under the existing regulatory framework?	☐ YES ☐ POTENTIALLY ☒ NO
P.9.12.1 |	risk of potential accidental or unintended introductions including the transportation of substrates and vectors (such as soil, ballast, and plant materials) that may harbour alien species.	☐ YES ☐ POTENTIALLY ☒ NO
P.9.12.2 |	risk of spreading alien species into areas in which they have not already been established?	☐ YES ☐ POTENTIALLY ☒ NO
If the answer is "yes" or "potentially" to any of the above question, please provide a brief description of the project situation below. Also, provide justification and/or evidence as necessary to demonstrate compliance with applicable requirements.		
Please add text here….		

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organisation name	ACE Energy Solutions Uganda Limited
Registration number with relevant authority	210632
Street/P.O. Box	Plot 27, Nankupa Road
Building	
City	Mbale
State/Region	
Postcode	
Country	Uganda
Telephone	
E-mail	
Website	
Contact person	Ruben Walker
Title	Director
Salutation	Mr
Last name	Walker
Middle name	
First name	Ruben
Department	ACE Uganda
Mobile	+256 784 587345
Direct tel.	
Personal e-mail	ruben@africancleanenergy.com

APPENDIX 3- LUF ADDITIONAL INFORMATION

Risk of change to the Project Area during Project Certification Period:	
Risk of change to the Project activities during Project Certification Period:	
Land-use history and current status of Project Area:	
Socio-Economic history:	
Forest management applied (past and future)	
Forest characteristics (including main tree species planted)	
Main social impacts (risks and benefits)	
Main environmental impacts (risks and benefits)	
Financial structure	
Infrastructure (roads/houses etc):	
Water bodies:	
Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:	
Where indigenous people and local communities are situated:	
Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:	

APPENDIX 4 - DESIGN CHANGES

A4.1. Details of proposed or actual design change

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A4.2. Describe the Impacts of Design Change on the following

a. ***Additionality***

>>

b. ***Applicability of methodology and other methodological regulatory documents with which the project activity has been certified***

>>

c. ***Compliance with the monitoring plan of the applied methodology***

>>

d. ***Level of accuracy and completeness in the monitoring of the project activity compared with the requirements contained in the registered monitoring plan***

>>

e. ***Scale of the project activity***

>>

f. ***Stakeholder consultation***

>>

g. ***Sustainable development criteria***

>>

h. ***Safeguarding Assessment***

>>

i. Compliance with applicable legislation

>>

Revision History

Version	Date	Remarks
2.3	Dd/mm/yyyy	Editorial changes in line with V2.1 of the Safeguarding Principles and Requirements
2.2	21 June 2023	Editorial changes in line with V2.0 of the Safeguarding Principles and Requirements
2.1	14 April 2023	Integrated the design change memo as annex of the document.
2.0	4 May 2022	
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption